# Early-Pandemic COVID-19 Epidemiologic Analysis

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```
library(ggplot2)
```

# 1 Study Data

This study uses a 10,000-record sample of reported COVID-19 cases restricted to the early pre-vaccine study period from March 15 through December 10, 2020.

The dataset for this study was acquired from the CDC COVID-19 Case Surveillance Public Use Data through the CDC Socrata API.

The study dataset is a computational subsample of and was constructed by randomly selecting 20 calendar dates within the study period using a fixed random seed (42). For each selected date, up to 500 records were retrieved from a contiguous block of surveillance records, with the

starting position selected using a date-specific random seed. Records were ordered by their internal Socrata identifier before retrieval.

The resulting dataset contains 10,000 surveillance records and 12 variables. Since sampling was performed by date and contiguous record blocks rather than by simple random sampling across all eligible cases, the sample should not be considered representative of all reported COVID-19 cases during the study period.

```
# Load COVID-19 surveillance sample
covid <- read.csv(
    "Data/covid_early_pandemic_sample_10k.csv",
    stringsAsFactors = FALSE
)

cat(sprintf("Rows: %s | Columns: %s\n",
    format(nrow(covid), big.mark = ","),
    format(ncol(covid), big.mark = ",")))
```

```
Rows: 10,000 | Columns: 12
```

## 2 Analytic Cohort

The analysis is restricted to cases with known exposure and outcome status. Exposure is defined by `medcond_yn`, indicating whether a pre-existing medical condition was reported. The outcome is defined by `hosp_yn`, indicating a hospitalization was reported.

`medcond_yn` and `hosp_yn` are coded as {`Yes`, `No`, `Unknown`, `Missing`}. Only `Yes` and `No` records for both variables were retained in the exposure-outcome complete-case cohort.

```
analysis <- covid[
    covid$medcond_yn %in% c("Yes", "No") &
    covid$hosp_yn %in% c("Yes", "No"),
]

cat(sprintf("Rows: %s | Columns: %s\n",
    format(nrow(analysis), big.mark = ","),
    format(ncol(analysis), big.mark = ",")))
```

```
Rows: 1,464 | Columns: 12
```

The resulting exposure-outcome complete-case cohort contained 1464 cases (14.64% of the original sample). The remaining 8536 cases (85.36%) were excluded because exposure status, hospitalization status, or both were not recorded as Yes or No.

## 2.1 Medical Condition Reporting Patterns

```
# Medical-condition reporting status
covid_medcond_yn <- table(
    covid$medcond_yn,
    useNA = "ifany"
)

# Convert frequency table to a data frame
medcond_reporting <- data.frame(
    Status = names(covid_medcond_yn),
    Count = as.integer(covid_medcond_yn),
    Percentage = sprintf(
        "%.2f%%",
        100 * as.numeric(covid_medcond_yn) / sum(covid_medcond_yn)
    )
)

knitr::kable(
    medcond_reporting,
    caption = "Medical Condition Reporting Status",
    align = c("l", "r", "r"),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 1: Medical Condition Reporting Status

| Status | Count | Percentage |
|---|---|---|
| Missing | 7358 | 73.58% |
| No | 973 | 9.73% |
| Unknown | 826 | 8.26% |
| Yes | 843 | 8.43% |

## 2.2 Hospitalization Reporting Patterns

```
# Hospitalization reporting status
covid_hosp_yn <- table(
    covid$hosp_yn,
    useNA = "ifany"
)

# Convert frequency table to a data frame
hosp_reporting <- data.frame(
    Status = names(covid_hosp_yn),
    Count = as.integer(covid_hosp_yn),
    Percentage = sprintf(
        "%.2f%%",
        100 * as.numeric(covid_hosp_yn) / sum(covid_hosp_yn)
    )
)

knitr::kable(
    hosp_reporting,
    caption = "Hospitalization Reporting Status",
    align = c("l", "r", "r"),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 2: Hospitalization Reporting Status

| Status | Count | Percentage |
|---|---|---|
| Missing | 3037 | 30.37% |
| No | 4720 | 47.20% |
| Unknown | 1220 | 12.20% |
| Yes | 1023 | 10.23% |

## 2.3 Joint Reporting Patterns

The joint distribution was examined to identify how frequently medical-condition and hospitalization status were reported together.

```r
# Examine joint reporting patterns
covid_joint <- table(
    covid$medcond_yn,
    covid$hosp_yn,
    useNA = "ifany"
)

knitr::kable(
    covid_joint,
    caption = "Joint Distribution of Medical Condition and Hospitalization Reporting Status",
    align = c("l", rep("r", ncol(covid_joint))),
    booktabs = TRUE,
    row.names = TRUE
)
```

Table 3: Joint Distribution of Medical Condition and Hospitalization Reporting Status

| | Missing | No | Unknown | Yes |
|---|---|---|---|---|
| Missing | 2838 | 3397 | 470 | 653 |
| No | 109 | 728 | 109 | 27 |
| Unknown | 43 | 136 | 554 | 93 |
| Yes | 47 | 459 | 87 | 250 |

```r
# Percentage with known medical-condition status
known_medical_condition <- mean(covid$medcond_yn %in% c("Yes", "No")) * 100

# Percentage with known hospitalization status
known_hospitalization <- mean(covid$hosp_yn %in% c("Yes", "No")) * 100

# Percentage with BOTH exposure and outcome known
known_both <- mean(
    covid$medcond_yn %in% c("Yes", "No") &
    covid$hosp_yn %in% c("Yes", "No")
) * 100

completeness_report <- data.frame(
    Metric = c(
        "Known Hospitalization Status",
        "Known Medical Condition Status",
        "Both Exposure and Outcome Known"
    ),
```

```
    Percentage = sprintf("%.2f%%", c(
        known_hospitalization,
        known_medical_condition,
        known_both
    ))
)

knitr::kable(
    completeness_report,
    caption = "Reporting Completeness",
    align = c("l", rep("r", ncol(completeness_report) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 4: Reporting Completeness

| Metric | Percentage |
|---|---|
| Known Hospitalization Status | 57.43% |
| Known Medical Condition Status | 18.16% |
| Both Exposure and Outcome Known | 14.64% |

Reporting completeness differed substantially between variables. Hospitalization status was known for 57.43% of sampled records, while pre-existing medical condition status was known for only 18.16%. Both variables were known for 14.64% of sampled records.

Restricting the analysis to cases with known exposure and outcome status may introduce selection bias if the excluded cases differ systematically from the included cases.

# 3 Crude Association

## 3.1 Exposure-Outcome Table

We first estimate the crude association between the exposure (reported pre-existing medical condition) and outcome (reported hospitalization).

```
# Construct 2x2 table with exposed/hospitalized first
exposure_outcome <- table(
    `Medical Condition (Exposure)` = factor(
```

```
        analysis$medcond_yn,
        levels = c("Yes", "No")
    ),
    `Hospitalized (Outcome)` = factor(
        analysis$hosp_yn,
        levels = c("Yes", "No")
    )
)

knitr::kable(
    exposure_outcome,
    caption = "Contingency Table",
    align = c("l", rep("r", ncol(exposure_outcome) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 5: Contingency Table

| Yes | No |
|---|---|
| 250 | 459 |
| 27 | 728 |

## 3.2 Risk Ratio and Risk Difference

Here $a$ and $b$ represent hospitalized and non-hospitalized exposed cases, respectively. Similarly, $c$ and $d$ represent hospitalized and non-hospitalized unexposed cases, respectively. The crude risk ratio (RR) and risk difference (RD) are defined as:

$$RR = \frac{a/(a+b)}{c/(c+d)} \qquad RD = \frac{a}{a+b} - \frac{c}{c+d}$$

Hospitalization risk is estimated separately among exposed and unexposed cases.

```
# Extract cells from exposure-outcome table
a <- exposure_outcome["Yes", "Yes"]
b <- exposure_outcome["Yes", "No"]
c <- exposure_outcome["No", "Yes"]
d <- exposure_outcome["No", "No"]
```

```r
# Hospitalization risks
risk_exposed <- a / (a + b)
risk_unexposed <- c / (c + d)

# Effect measures
risk_ratio <- risk_exposed / risk_unexposed
risk_difference <- risk_exposed - risk_unexposed

# Display hospitalization risks and effect measures
results <- data.frame(
    Measure = c(
        "Risk in Exposed (reported medical condition)",
        "Risk in Unexposed (no reported medical condition)",
        "Risk Ratio (RR)",
        "Risk Difference (RD)"
    ),
    Value = c(
        sprintf("%.3f (%.1f%%)", risk_exposed, risk_exposed * 100),
        sprintf("%.3f (%.1f%%)", risk_unexposed, risk_unexposed * 100),
        sprintf("%.3f", risk_ratio),
        sprintf("%.3f (%+.1f pp)", risk_difference, risk_difference * 100)
    )
)

knitr::kable(
    results,
    caption = "Risk Estimates & Measures of Effect",
    align = c("l", rep("r", ncol(results) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 6: Risk Estimates & Measures of Effect

| Measure | Value |
|---|---|
| Risk in Exposed (reported medical condition) | 0.353 (35.3%) |
| Risk in Unexposed (no reported medical condition) | 0.036 (3.6%) |
| Risk Ratio (RR) | 9.860 |
| Risk Difference (RD) | 0.317 (+31.7 pp) |

Among cases in the complete-case analytic cohort, hospitalization was reported for 35.3% of

individuals with a reported pre-existing medical condition and 3.6% of individuals without one. The crude risk ratio was 9.86, indicating that the proportion hospitalized was approximately 9.86 times as high among individuals with reported medical conditions. The crude risk difference was 31.7 percentage points, representing the absolute difference in hospitalization proportions between the two groups.

### 3.3 95% Confidence Interval for Risk Ratio

A 95% confidence interval (CI) for the crude risk ratio was calculated using the normal approximation to the log-transformed risk ratio. The standard error of the log risk ratio is estimated as:

$$SE[\ln(RR)] = \sqrt{\frac{1}{a} - \frac{1}{a+b} + \frac{1}{c} - \frac{1}{c+d}}$$

The confidence interval is calculated on the logarithmic scale and then exponentiated to return to the risk-ratio scale.

$$95\%CI_{RR} = \exp[\ln(RR) \pm 1.96 \times SE[\ln(RR)]]$$

```
# Standard error of the log risk ratio
se_log_rr <- sqrt(
    (1 / a) - (1 / (a + b)) +
    (1 / c) - (1 / (c + d))
)

# Calculate confidence limits on log scale
log_rr <- log(risk_ratio)

log_rr_lower <- log_rr - (1.96 * se_log_rr)
log_rr_upper <- log_rr + (1.96 * se_log_rr)

# Transform confidence limits back to risk-ratio scale
rr_lower <- exp(log_rr_lower)
rr_upper <- exp(log_rr_upper)

rr_summary <- data.frame(
    Estimate = sprintf("%.3f", risk_ratio),
    `95% CI Lower` = sprintf("%.3f", rr_lower),
    `95% CI Upper` = sprintf("%.3f", rr_upper),
    check.names = FALSE
```

```
)

knitr::kable(
    rr_summary,
    caption = "Risk Ratio (95% Confidence Interval)",
    align = c("l", rep("r", ncol(rr_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 7: Risk Ratio (95% Confidence Interval)

| Estimate | 95% CI Lower | 95% CI Upper |
|---|---|---|
| 9.860 | 6.719 | 14.470 |

The crude risk ratio was 9.86 (95% CI: 6.72–14.47). The confidence interval excludes the null value of 1, indicating evidence of a positive crude association between reported pre-existing medical condition and hospitalization.

### 3.4 95% Confidence Interval for Risk Difference

A 95% confidence interval for the crude risk difference was calculated using the normal approximation for the difference between two independent proportions. The standard error is estimated as:

$$SE(RD) = \sqrt{\frac{R_1(1-R_1)}{n_1} + \frac{R_0(1-R_0)}{n_0}}$$

Here, $R_1$ and $R_0$ represent the observed hospitalization proportions among exposed and unexposed cases, respectively, while $n_1$ and $n_0$ represent the corresponding group sizes.

The confidence interval is calculated directly on the risk-difference scale:

$$95\%CI_{RD} = RD \pm 1.96 \times SE(RD)$$

```r
# Group sample sizes
n_exposed <- a + b
n_unexposed <- c + d

# Standard error of risk difference
se_rd <- sqrt(
    (risk_exposed * (1 - risk_exposed) / n_exposed) +
    (risk_unexposed * (1 - risk_unexposed) / n_unexposed)
)

# Calculate confidence limits
rd_lower <- risk_difference - (1.96 * se_rd)
rd_upper <- risk_difference + (1.96 * se_rd)

rd_summary <- data.frame(
  Scale = c("Proportion", "Percentage Points (pp)"),
  Estimate = c(
    sprintf("%+.3f", risk_difference),
    sprintf("%+.2f pp", risk_difference * 100)
  ),
  `95% CI Lower` = c(
    sprintf("%+.3f", rd_lower),
    sprintf("%+.2f pp", rd_lower * 100)
  ),
  `95% CI Upper` = c(
    sprintf("%+.3f", rd_upper),
    sprintf("%+.2f pp", rd_upper * 100)
  ),
  check.names = FALSE
)

knitr::kable(
    rd_summary,
    caption = "Risk Difference (95% Confidence Interval)",
    align = c("l", rep("r", ncol(rd_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 8: Risk Difference (95% Confidence Interval)

| Scale | Estimate | 95% CI Lower | 95% CI Upper |
|---|---|---|---|
| Proportion | +0.317 | +0.279 | +0.354 |
| Percentage Points (pp) | +31.68 pp | +27.93 pp | +35.44 pp |

The crude risk difference was 31.7 percentage points (95% CI: 27.9–35.4 percentage points). The confidence interval excludes the null value of 0, indicating evidence of a positive crude difference in hospitalization proportions between reported pre-existing medical condition and hospitalization.

These estimates describe crude, unadjusted associations and do not account for potential confounding by age or sex. Both confidence intervals were calculated using the normal approximation, which may be inaccurate for small sample sizes or extreme proportions.

# 4 Descriptive Analysis

Age and sex were hypothesized to be potential confounders based on the directed acyclic graph (DAG) and prior literature review. Before fitting adjusted models, their distributions were examined across exposure groups, along with the observed proportion hospitalized within each age and sex category.

## 4.1 Age

### 4.1.1 Age Distribution by Exposure

The age distribution was examined separately among cases with and without reported pre-existing medical conditions.

```
# Age distribution by exposure status
age_exposure <- table(
    `Age Group` = analysis$age_group,
    `Medical Condition` = analysis$medcond_yn
)

# Calculate column percentages (within exposure group)
age_exposure_prop <- prop.table(age_exposure, margin = 2) * 100

cat("Age Distribution by Exposure Status\n")
```

Age Distribution by Exposure Status

```
knitr::kable(
    age_exposure,
    caption = "1. Case Counts (with Margins)",
    align = c("l", rep("r", ncol(age_exposure) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 9: 1. Case Counts (with Margins)

| No | Yes |
|---|---|
| 134 | 15 |
| 65 | 23 |
| 171 | 78 |
| 164 | 98 |
| 52 | 46 |
| 104 | 92 |
| 51 | 224 |
| 9 | 49 |
| 5 | 84 |

```
# Combine count and % into a single clean string: "n (pct%)"
age_pct_formatted <- matrix(
  sprintf("%.1f%%", age_exposure_prop),
  nrow = nrow(age_exposure),
  dimnames = dimnames(age_exposure)
)

knitr::kable(
    age_pct_formatted,
    caption = "2. Column Percentages (% within exposure group)",
    align = c("l", rep("r", ncol(age_pct_formatted) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 10: 2. Column Percentages (% within exposure group)

| No | Yes |
|---|---|
| 17.7% | 2.1% |
| 8.6% | 3.2% |
| 22.6% | 11.0% |
| 21.7% | 13.8% |
| 6.9% | 6.5% |
| 13.8% | 13.0% |
| 6.8% | 31.6% |
| 1.2% | 6.9% |
| 0.7% | 11.8% |

Column percentages represent the age distribution within each exposure group, $P(Age \mid E)$. Each exposure column sums to 100%.

```
# Age groups aged 60+
older_groups <- c(
    "60 - 69 Years",
    "70 - 79 Years",
    "80+ Years"
)

# Case counts and totals
n_older <- colSums(age_exposure[older_groups, , drop = FALSE])
n_total <- colSums(age_exposure)

# Percentage  aged 60+ in each exposure group
older_percent <- (n_older / n_total) * 100

older_summary <- data.frame(
  `Medical Condition` = names(older_percent),
  `Aged 60+ (n)` = format(n_older, big.mark = ","),
  `Total (N)` = format(n_total, big.mark = ","),
  `Percentage Aged 60+` = sprintf("%.1f%%", older_percent),
  check.names = FALSE
)

knitr::kable(
    older_summary,
    caption = "Proportion Aged 60+ by Exposure Status",
```

```
    align = c("l", rep("r", ncol(older_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 11: Proportion Aged 60+ by Exposure Status

| Medical Condition | Aged 60+ (n) | Total (N) | Percentage Aged 60+ |
|---|---|---|---|
| No | 65 | 755 | 8.6% |
| Yes | 357 | 709 | 50.4% |

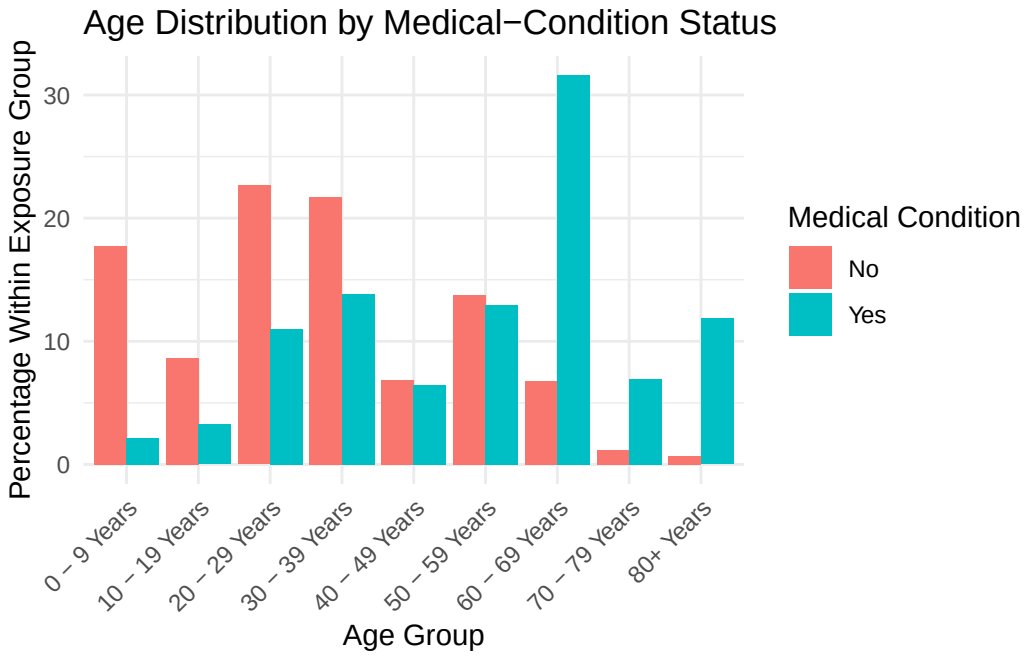
The age distribution differed substantially between exposure groups. Among cases with reported pre-existing medical conditions, 50.4% were aged 60 years or older, compared with 8.6% among cases without reported conditions. Older individuals were disproportionately represented in the exposed group, suggesting that age may contribute to the observed crude association between reported pre-existing medical condition and hospitalization.

```
# Convert column percentages to a data frame for plotting
age_exposure_df <- as.data.frame(
    prop.table(age_exposure, margin = 2) * 100
)

names(age_exposure_df) <- c(
    "Age",
    "Medical_Condition",
    "Percentage"
)

# Plot age distribution within each exposure group
ggplot(
    age_exposure_df,
    aes(
        x = Age,
        y = Percentage,
        fill = Medical_Condition
    )
) +
    geom_col(position = "dodge") +
    labs(
        title = "Age Distribution by Medical-Condition Status",
        x = "Age Group",
```

```
        y = "Percentage Within Exposure Group",
        fill = "Medical Condition"
    ) +
    theme_minimal() +
    theme(
        axis.text.x = element_text(angle = 45, hjust = 1)
    )
```




### 4.1.2 Hospitalization by Age

The proportion hospitalized was calculated within each age group to examine whether observed hospitalization proportion varies by age.

```
# Hospitalization status by age group
age_outcome <- table(
    `Age Group` = analysis$age_group,
    `Hospitalized` = analysis$hosp_yn
)

# Row percentages: hospitalization status within each age group
age_outcome_pct <- prop.table(age_outcome, margin = 1) * 100

cat("Hospitalization Distribution by Age Group\n")
```

Hospitalization Distribution by Age Group

```
knitr::kable(
    age_outcome,
    caption = "1. Case Counts (with Row Margins)",
    align = c("l", rep("r", ncol(age_outcome) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 12: 1. Case Counts (with Row Margins)

| No | Yes |
|---|---|
| 145 | 4 |
| 86 | 2 |
| 240 | 9 |
| 238 | 24 |
| 87 | 11 |
| 184 | 12 |
| 127 | 148 |
| 27 | 31 |
| 53 | 36 |

```
age_pct_formatted <- matrix(
  sprintf("%.1f%%", age_outcome_pct),
  nrow = nrow(age_outcome),
  dimnames = dimnames(age_outcome)
)

knitr::kable(
    age_pct_formatted,
    caption = "2. Row Percentages (% within age group, summing across rows)",
    align = c("l", rep("r", ncol(age_pct_formatted) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 13: 2. Row Percentages (% within age group, summing across rows)

| No | Yes |
|---|---|
| 97.3% | 2.7% |
| 97.7% | 2.3% |
| 96.4% | 3.6% |
| 90.8% | 9.2% |
| 88.8% | 11.2% |
| 93.9% | 6.1% |
| 46.2% | 53.8% |
| 46.6% | 53.4% |
| 59.6% | 40.4% |

Row percentages represent hospitalization status within each age group, $P(Y \mid Age)$. Each age-group row sums to 100%.

```
# Calculate percentage hospitalized within each age group
hosp_yes_count <- age_outcome[, "Yes"]
total_in_age <- rowSums(age_outcome)
age_hosp_percent <- (hosp_yes_count / total_in_age) * 100

hosp_risk_summary <- data.frame(
  `Age Group` = names(age_hosp_percent),
  `Hospitalized (n)` = format(hosp_yes_count, big.mark = ","),
  `Total Cases (N)` = format(total_in_age, big.mark = ","),
  `Hospitalization Risk` = sprintf("%.1f%%", age_hosp_percent),
  check.names = FALSE
)

knitr::kable(
    hosp_risk_summary,
    caption = "Risk of Hospitalization by Age Group",
    align = c("l", rep("r", ncol(hosp_risk_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 14: Risk of Hospitalization by Age Group

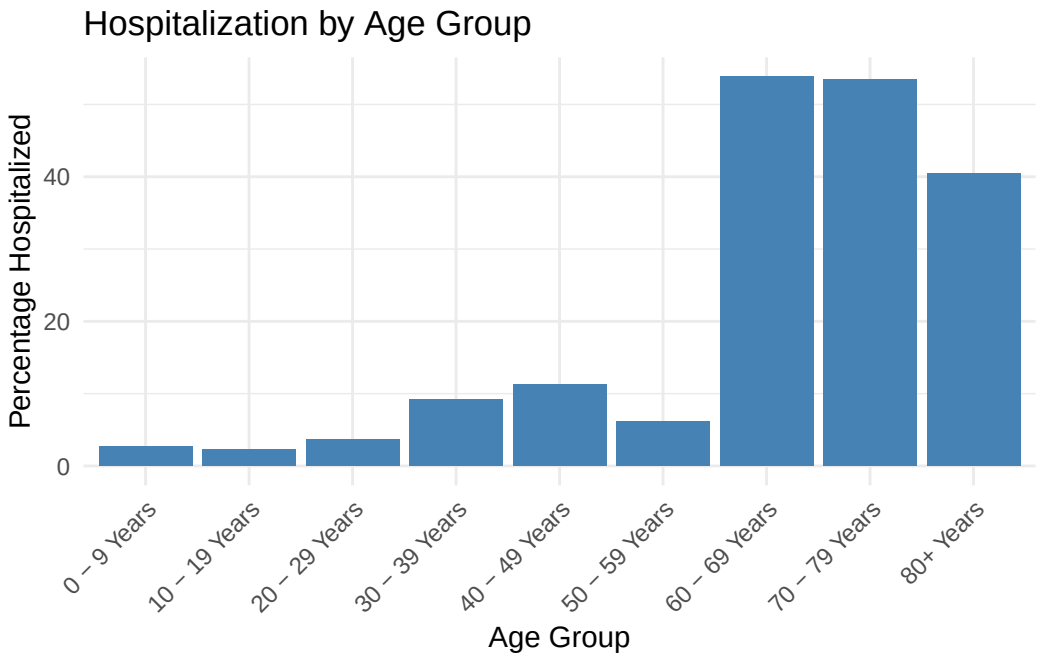
| Age Group | Hospitalized (n) | Total Cases (N) | Hospitalization Risk |
|---|---|---|---|
| 0 - 9 Years | 4 | 149 | 2.7% |
| 10 - 19 Years | 2 | 88 | 2.3% |
| 20 - 29 Years | 9 | 249 | 3.6% |
| 30 - 39 Years | 24 | 262 | 9.2% |
| 40 - 49 Years | 11 | 98 | 11.2% |
| 50 - 59 Years | 12 | 196 | 6.1% |
| 60 - 69 Years | 148 | 275 | 53.8% |
| 70 - 79 Years | 31 | 58 | 53.4% |
| 80+ Years | 36 | 89 | 40.4% |

Hospitalization proportions varied substantially across age groups. Among cases aged 60–69 years, 53.8% were hospitalized, compared with 3.6% among cases aged 20–29 years. The proportion hospitalized among cases aged 80 years or older was 40.4%, indicating that the observed pattern did not increase consistently across all age categories. These findings suggest a strong descriptive association between age and hospitalization within the analytic cohort.

```
# Convert hospitalization percentages into a data frame for plotting
age_hosp_df <- data.frame(
    Age = names(age_hosp_percent),
    Percentage = as.numeric(age_hosp_percent)
)

# Plot hospitalization percentage within each age group
ggplot(
    age_hosp_df,
    aes(
        x = Age,
        y = Percentage
    )
) +
    geom_col(fill = "steelblue") +
    labs(
        title = "Hospitalization by Age Group",
        x = "Age Group",
        y = "Percentage Hospitalized"
    ) +
    theme_minimal() +
    theme(
```

```
        axis.text.x = element_text(angle = 45, hjust = 1)
    )
```




## 4.2 Sex

### 4.2.1 Sex by Exposure

The sex distribution was examined separately among cases with and without reported pre-existing medical conditions.

```
# Sex distribution by exposure status
sex_exposure <- table(
    `Sex` = analysis$sex,
    `Medical Condition` = analysis$medcond_yn
)

# Calculate column percentages (within exposure group)
sex_exposure_prop <- prop.table(sex_exposure, margin = 2) * 100

cat("Sex Distribution by Exposure Status\n")
```

```
Sex Distribution by Exposure Status
```

```r
knitr::kable(
    sex_exposure,
    caption = "1. Case Counts (with Margins)",
    align = c("l", rep("r", ncol(sex_exposure) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 15: 1. Case Counts (with Margins)

| No | Yes |
|---|---|
| 368 | 327 |
| 386 | 381 |
| 1 | 0 |
| 0 | 1 |

```r
# Format percentages
sex_pct_formatted <- matrix(
    sprintf("%.1f%%", sex_exposure_prop),
    nrow = nrow(sex_exposure),
    dimnames = dimnames(sex_exposure)
)

knitr::kable(
    sex_pct_formatted,
    caption = "2. Column Percentages (% within exposure group)",
    align = c("l", rep("r", ncol(sex_pct_formatted) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 16: 2. Column Percentages (% within exposure group)

| No | Yes |
|---|---|
| 48.7% | 46.1% |
| 51.1% | 53.7% |
| 0.1% | 0.0% |
| 0.0% | 0.1% |

Column percentages represent the sex distribution within each exposure group, $P(Sex \mid E)$. Each exposure column sums to 100%.

```r
# Calculate percentage of males within each exposure group
male_count <- sex_exposure["Male", ]
total_by_exposure <- colSums(sex_exposure)
male_percent <- (male_count / total_by_exposure) * 100

male_summary <- data.frame(
    `Medical Condition` = names(male_percent),
    `Male (n)` = format(male_count, big.mark = ","),
    `Total (N)` = format(total_by_exposure, big.mark = ","),
    `Percentage Male` = sprintf("%.1f%%", male_percent),
    check.names = FALSE
)

knitr::kable(
    male_summary,
    caption = "Proportion Male by Exposure Status",
    align = c("l", rep("r", ncol(male_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 17: Proportion Male by Exposure Status

| Medical Condition | Male (n) | Total (N) | Percentage Male |
|---|---|---|---|
| No | 386 | 755 | 51.1% |
| Yes | 381 | 709 | 53.7% |

The sex distribution was similar between exposure groups. Males accounted for 53.7% of cases with reported pre-existing medical conditions, compared with 51.1% of cases without reported conditions. These relatively small differences suggest that sex may contribute less to the crude exposure-outcome association than age.

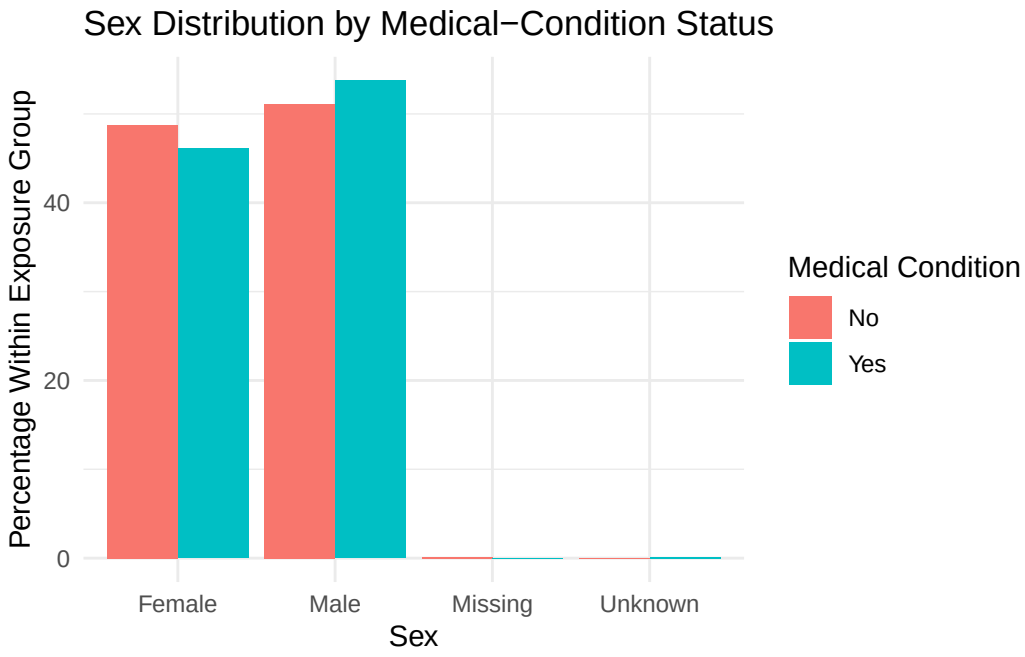
```r
# Convert column percentages to dataframe for plotting
sex_exposure_df <- as.data.frame(
    prop.table(sex_exposure, margin = 2) * 100
)

sex_exposure_df <- as.data.frame(sex_exposure_prop)

names(sex_exposure_df) <- c(
    "Sex",
```

```
    "Medical_Condition",
    "Percentage"
)

# Plot sex distribution within each exposure group
ggplot(
    sex_exposure_df,
    aes(
        x = Sex,
        y = Percentage,
        fill = Medical_Condition
    )
) +
    geom_col(position = "dodge") +
    labs(
        title = "Sex Distribution by Medical-Condition Status",
        x = "Sex",
        y = "Percentage Within Exposure Group",
        fill = "Medical Condition"
    ) +
    theme_minimal()
```



### 4.2.2 Hospitalization by Sex

The proportion hospitalized was calculated within each sex category to examine differences in hospitalization reports between males and females.

```
# Hospitalization status by sex
sex_outcome <- table(
    `Sex` = analysis$sex,
    `Hospitalized` = analysis$hosp_yn
)

# Calculate row percentages (within sex category)
sex_outcome_pct <- prop.table(sex_outcome, margin = 1) * 100

cat("Hospitalization Distribution by Sex\n")
```

```
Hospitalization Distribution by Sex
```

```
knitr::kable(
    sex_outcome,
    caption = "1. Case Counts (with Row Margins)",
    align = c("l", rep("r", ncol(sex_outcome) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 18: 1. Case Counts (with Row Margins)

| No | Yes |
|---|---|
| 573 | 122 |
| 612 | 155 |
| 1 | 0 |
| 1 | 0 |

```
# Format percentages for presentation
sex_outcome_formatted <- matrix(
    sprintf("%.1f%%", sex_outcome_pct),
    nrow = nrow(sex_outcome),
    dimnames = dimnames(sex_outcome)
)
```

```
knitr::kable(
    sex_outcome_formatted,
    caption = "2. Row Percentages (% within sex category)",
    align = c("l", rep("r", ncol(sex_outcome_formatted) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 19: 2. Row Percentages (% within sex category)

| No | Yes |
|---|---|
| 82.4% | 17.6% |
| 79.8% | 20.2% |
| 100.0% | 0.0% |
| 100.0% | 0.0% |

Row percentages represent hospitalization status within each sex category, $P(Y \mid Sex)$. Each sex row sums to 100%.

```
# Calculate percentage hospitalized within each sex category
hosp_yes_by_sex <- sex_outcome[, "Yes"]
total_by_sex <- rowSums(sex_outcome)
sex_hosp_percent <- (hosp_yes_by_sex / total_by_sex) * 100

sex_hosp_summary <- data.frame(
    `Sex` = names(sex_hosp_percent),
    `Hospitalized (n)` = format(hosp_yes_by_sex, big.mark = ","),
    `Total Cases (N)` = format(total_by_sex, big.mark = ","),
    `Hospitalization Proportion` = sprintf("%.1f%%", sex_hosp_percent),
    check.names = FALSE
)

knitr::kable(
    sex_hosp_summary,
    caption = "Hospitalization Proportion by Sex",
    align = c("l", rep("r", ncol(sex_hosp_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 20: Hospitalization Proportion by Sex

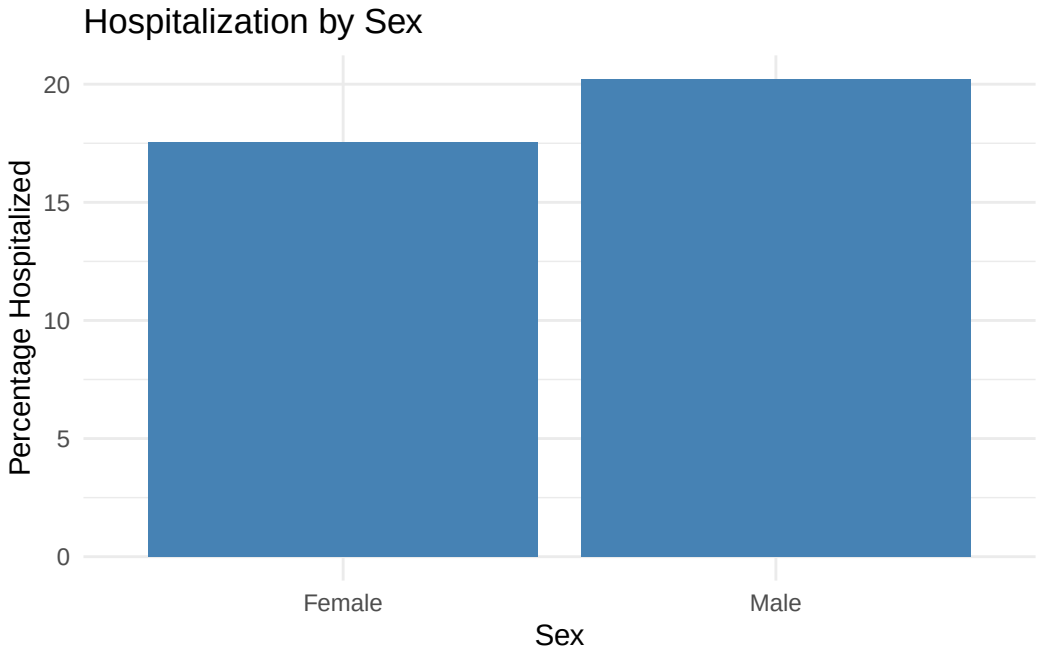
| Sex | Hospitalized (n) | Total Cases (N) | Hospitalization Proportion |
|---|---|---|---|
| Female | 122 | 695 | 17.6% |
| Male | 155 | 767 | 20.2% |
| Missing | 0 | 1 | 0.0% |
| Unknown | 0 | 1 | 0.0% |

Hospitalization proportions differed modestly between males and females. Among male cases, 20.2% were hospitalized, compared with 17.6% among female cases. This suggests a relatively small descriptive association between sex and hospitalization within the analytic cohort.

```
# Convert hospitalization percentages into dataframe for plotting
sex_hosp_df <- data.frame(
    Sex = names(sex_hosp_percent),
    Percentage = as.numeric(sex_hosp_percent)
)

# Keep Female and Male categories for visualization
sex_hosp_df <- subset(
    sex_hosp_df,
    Sex %in% c("Female", "Male")
)

# Plot hospitalization percentage within each sex category
ggplot(
    sex_hosp_df,
    aes(
        x = Sex,
        y = Percentage
    )
) +
    geom_col(fill = "steelblue") +
    labs(
        title = "Hospitalization by Sex",
        x = "Sex",
        y = "Percentage Hospitalized"
    ) +
    theme_minimal()
```




### 4.3 Summary of Potential Confounders

The descriptive comparisons were used to examine the relationships of age and sex with both reported pre-existing medical conditions and hospitalization. Differences in these distributions may contribute to the observed crude association.

Age and sex were retained as adjustment variables based on the hypothesized causal structure and prior literature review rather than solely on the descriptive results. The following adjusted analysis examines the exposure-outcome association while accounting for these variables.

## 5 Adjusted Analysis

Age and sex were identified as potential confounders based on the hypothesized causal structure and prior literature review. To examine the association between reported pre-existing medical conditions and hospitalization while accounting for these variables, a multivariable logistic regression model was fitted.

### 5.1 Adjusted Analytic Cohort

The adjusted analysis was restricted to cases with known exposure, outcome, age, and sex status. Cases with sex recorded as `Missing` or `Unknown`, or without a valid age group, were excluded from the adjusted analytic cohort.

```r
# Original exposure-outcome complete-case cohort
n_initial <- nrow(analysis)

# Restrict to valid age and sex categories
adjusted_analysis <- analysis[
    analysis$sex %in% c("Female", "Male") &
    !is.na(analysis$age_group) &
    analysis$age_group %in% c(
        "0 - 9 Years",
        "10 - 19 Years",
        "20 - 29 Years",
        "30 - 39 Years",
        "40 - 49 Years",
        "50 - 59 Years",
        "60 - 69 Years",
        "70 - 79 Years",
        "80+ Years"
    ),
    ,
    drop = FALSE
]

# Calculate cohort retention
n_adjusted <- nrow(adjusted_analysis)
n_excluded <- n_initial - n_adjusted
retention_pct <- (n_adjusted / n_initial) * 100

cohort_summary <- data.frame(
    `Cohort` = c(
        "Exposure-Outcome Complete Cases",
        "Excluded for Age or Sex",
        "Adjusted Analytic Cohort"
    ),
    `Cases (n)` = format(
        c(n_initial, n_excluded, n_adjusted),
        big.mark = ","
    ),
    check.names = FALSE
)

knitr::kable(
    cohort_summary,
```

```
    caption = "Adjusted Analytic Cohort Summary",
    align = c("l", rep("r", ncol(cohort_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 21: Adjusted Analytic Cohort Summary

| Cohort | Cases (n) |
|---|---|
| Exposure-Outcome Complete Cases | 1,464 |
| Excluded for Age or Sex | 2 |
| Adjusted Analytic Cohort | 1,462 |

```
cat(
    "Retention Rate:",
    sprintf("%.1f%%", retention_pct)
)
```

```
Retention Rate: 99.9%
```

Of the 1,464 cases in the exposure-outcome complete-case cohort, 2 were excluded due to missing, unknown, or invalid age or sex information. The resulting adjusted analytic cohort contained 1,462 cases, representing 99.9% of the original complete-case cohort.

### 5.1.1 Variable Encoding and Reference Categories

The outcome, exposure, and adjustment variables were encoded as categorical factors to define the reference categories used in the logistic regression model

```
# Hospitalization: No is the reference, Yes is the event
adjusted_analysis$hospitalized <- factor(
    adjusted_analysis$hosp_yn,
    levels = c("No", "Yes")
)

# Medical condition: No is the reference
adjusted_analysis$medcond <- factor(
    adjusted_analysis$medcond_yn,
    levels = c("No", "Yes")
```

```
)

# Sex: Female is the reference
adjusted_analysis$sex_factor <- factor(
    adjusted_analysis$sex,
    levels = c("Female", "Male")
)

# Age: 0--9 years is the reference
adjusted_analysis$age_factor <- factor(
    adjusted_analysis$age_group,
    levels = c(
        "0 - 9 Years",
        "10 - 19 Years",
        "20 - 29 Years",
        "30 - 39 Years",
        "40 - 49 Years",
        "50 - 59 Years",
        "60 - 69 Years",
        "70 - 79 Years",
        "80+ Years"
    )
)

# Verify reference categories and factor levels
model_variables <- adjusted_analysis[
    c("hospitalized", "medcond", "age_factor", "sex_factor")
]

reference_summary <- data.frame(
    `Variable` = names(model_variables),
    `Reference Category` = vapply(
        model_variables,
        function(x) levels(x)[1],
        character(1)
    ),
    `Number of Levels` = vapply(
        model_variables,
        nlevels,
        integer(1)
    ),
    check.names = FALSE
```

```
)

knitr::kable(
    reference_summary,
    caption = "Model Variable Reference Categories",
    align = c("l", rep("r", ncol(reference_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 22: Model Variable Reference Categories

| Variable | Reference Category | Number of Levels |
|---|---|---|
| hospitalized | No | 2 |
| medcond | No | 2 |
| age_factor | 0 - 9 Years | 9 |
| sex_factor | Female | 2 |

Hospitalization was modeled with `Yes` as the event and `No` as the reference category. Reported pre-existing medical-condition status used `No` as the reference category. Sex used `Female` as the reference category, while age used `0--9 Years` as the reference group.

## 5.2 Logistic Regression

A multivariable logistic regression model was fitted to examine the association between reported pre-existing medical conditions and hospitalization while adjusting for age group and sex. Hospitalization was modeled as a binary outcome, with medical-condition status as the primary exposure and age group and sex included as categorical covariates.

### 5.2.1 Model Specification

The logistic regression model expresses the log-odds of hospitalization as a linear combination of the exposure and adjustment variables:

$$\log(\frac{p_i}{1-p_i}) = \beta_0 + \beta_E E_i + \sum_{k=1}^{8} \beta A_k A_{ik} + \beta_S S_i,$$

where

$$p_i = P(Y_i = 1 \mid E_i, A_i, S_i).$$

In the equation above, $Y_i$ indicates hospitalization for case $i$, $E_i$ represents reported pre-existing medical-condition status, $A_{ik}$ represents the eight indicator variables corresponding to non-reference age groups, and $S_i$ represents sex.

The exposure coefficient $\beta_E$ represents the difference in conditional log-odds of hospitalization between cases with and without reported pre-existing medical conditions, holding age group and sex constant.

### 5.2.2 Model Estimation

```
# Fit logistic regression adjusting for age and sex
logistic_model <- glm(
    hospitalized ~ medcond + age_factor + sex_factor,
    data = adjusted_analysis,
    family = binomial(link = "logit")
)

# Extract model coefficient summary
model_summary <- summary(logistic_model)$coefficients

# Construct formatted regression coefficient table
coefficient_table <- data.frame(
    `Term` = rownames(model_summary),
    `Log-Odds Coefficient` = sprintf("%.3f", model_summary[, "Estimate"]),
    `Standard Error` = sprintf("%.3f", model_summary[, "Std. Error"]),
    `z-Statistic` = sprintf("%.3f", model_summary[, "z value"]),
    `p Value` = format.pval(
        model_summary[, "Pr(>|z|)"],
        digits = 3,
        eps = 0.001
    ),
    check.names = FALSE
)

knitr::kable(
    coefficient_table,
    caption = "Multivariable Logistic Regression Coefficients",
    align = c("l", rep("r", ncol(coefficient_table) - 1)),
```

```
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 23: Multivariable Logistic Regression Coefficients

| Term | Log-Odds Coefficient | Standard Error | z-Statistic | p Value |
|---|---|---|---|---|
| (Intercept) | -4.180 | 0.535 | -7.807 | <0.001 |
| medcondYes | 1.874 | 0.232 | 8.079 | <0.001 |
| age_factor10 - 19 Years | -0.527 | 0.893 | -0.590 | 0.555 |
| age_factor20 - 29 Years | -0.162 | 0.627 | -0.258 | 0.797 |
| age_factor30 - 39 Years | 0.615 | 0.567 | 1.084 | 0.278 |
| age_factor40 - 49 Years | 0.733 | 0.620 | 1.183 | 0.237 |
| age_factor50 - 59 Years | 0.000 | 0.607 | 0.000 | 1.000 |
| age_factor60 - 69 Years | 2.624 | 0.542 | 4.838 | <0.001 |
| age_factor70 - 79 Years | 2.567 | 0.598 | 4.295 | <0.001 |
| age_factor80+ Years | 1.964 | 0.585 | 3.358 | <0.001 |
| sex_factorMale | 0.281 | 0.182 | 1.538 | 0.124 |

The regression coefficients are estimated on the log-odds scale. Positive coefficients indicate higher conditional odds of hospitalization relative to the corresponding reference category, while negative coefficients indicate lower conditional odds. The coefficients for age group and sex describe associations conditional on the other variables included in the model.

### 5.2.3 Exposure Coefficient and Adjusted Odds Ratio

The coefficient for reported pre-existing medical conditions was extracted from the fitted model. Exponentiating this coefficient converts the estimated log-odds ratio to an adjusted odds ratio (OR):

$$OR_{\text{adj}} = \exp(\hat{\beta}_E).$$

A 95% Wald confidence interval was calculated using the standard error of the exposure coefficient. The confidence limits were first constructed on the log-odds scale and then exponentiated:

$$CI_{95\%}(OR_{\text{adj}}) = \left[\exp\left(\hat{\beta}_E - 1.96\, SE(\hat{\beta}_E)\right), \exp\left(\hat{\beta}_E + 1.96\, SE(\hat{\beta}_E)\right)\right].$$

```r
# Identify exposure coefficient
exposure_term <- "medcondYes"

# Extract coefficient and standard error
exposure_beta <- unname(
    coef(logistic_model)[exposure_term]
)

exposure_se <- summary(logistic_model)$coefficients[
    exposure_term,
    "Std. Error"
]

# Calculate adjusted odds ratio
adjusted_or <- exp(exposure_beta)

# Calculate 95% Wald confidence interval
or_ci_lower <- exp(exposure_beta - 1.96 * exposure_se)
or_ci_upper <- exp(exposure_beta + 1.96 * exposure_se)

or_summary <- data.frame(
    `Measure` = "Adjusted OR",
    `Log-Odds Coefficient` = sprintf("%.3f", exposure_beta),
    `Standard Error` = sprintf("%.3f", exposure_se),
    `Adjusted OR` = sprintf("%.3f", adjusted_or),
    `95% CI Lower` = sprintf("%.3f", or_ci_lower),
    `95% CI Upper` = sprintf("%.3f", or_ci_upper),
    check.names = FALSE
)

knitr::kable(
    or_summary,
    caption = "Adjusted Association: Medical Conditions and Hospitalization",
    align = c("l", rep("r", ncol(or_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 24: Adjusted Association: Medical Conditions and Hospitalization

| Measure | Log-Odds Coefficient | Standard Error | Adjusted OR | 95% CI Lower | 95% CI Upper |
|---|---|---|---|---|---|
| Adjusted OR | 1.874 | 0.232 | 6.513 | 4.134 | 10.260 |

#### 5.2.4 Interpretation of Logistic Regression

The estimated medical-condition coefficient was 1.874 on the log-odds scale. Exponentiating this coefficient yielded an adjusted odds ratio of 6.51 (95% CI: 4.13–10.26).

After accounting for age group and sex, cases with reported pre-existing medical conditions had approximately 6.51 times the odds of hospitalization compared with cases without reported conditions.

### 5.3 Regression Standardization

Regression standardization was used to estimate hospitalization proportions for the two exposure scenarios while accounting for the age and sex distribution of the adjusted-analysis cohort.

The fitted logistic regression model was used to predict hospitalization probabilities after setting reported pre-existing medical-condition status to `Yes` for every case and, separately, to `No` for every case. Each case's observed age group and sex were retained in both scenarios.

The predicted probabilities were then averaged across the same analytic cohort, producing two hospitalization proportions standardized to a common age and sex distribution.

#### 5.3.1 Constructing Exposure Scenarios

Two copies of the adjusted-analysis cohort were created. Pre-existing medical condition status was set to `Yes` for every case in one copy and `No` for every case in the other copy. Age and sex were left unchanged in both copies.

```
# Create two copies of the adjusted cohort
everyone_exposed <- adjusted_analysis
everyone_unexposed <- adjusted_analysis

# Assign reported medical conditions to every case
everyone_exposed$medcond <- factor(
    rep("Yes", nrow(everyone_exposed)),
```

```
    levels = levels(adjusted_analysis$medcond)
)

# Assign no reported medical conditions to every case
everyone_unexposed$medcond <- factor(
    rep("No", nrow(everyone_unexposed)),
    levels = levels(adjusted_analysis$medcond)
)

# Summarize the two exposure scenarios
scenario_summary <- data.frame(
    `Scenario` = c(
        "All Cases Exposed",
        "All Cases Unexposed"
    ),
    `Medical Condition` = c("Yes", "No"),
    `Cases (N)` = format(
        rep(nrow(adjusted_analysis), 2),
        big.mark = ","
    ),
    check.names = FALSE
)

knitr::kable(
    scenario_summary,
    caption = "Regression Standardization: Exposure Scenarios",
    align = c("l", rep("r", ncol(scenario_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 25: Regression Standardization: Exposure Scenarios

| Scenario | Medical Condition | Cases (N) |
|---|---|---|
| All Cases Exposed | Yes | 1,462 |
| All Cases Unexposed | No | 1,462 |

Both scenarios contain 1,462 cases with identical age and sex distributions. The only difference between the two datasets is the assigned medical-condition status. They do not represent actual observed data, but rather hypothetical situations in which every case is assigned the same exposure status while retaining their observed age and sex.

### 5.3.2 Predicting Standardized Hospitalization Proportions

The fitted logistic regression model was used to predict hospitalization probabilities for each case under both exposure scenarios.

*Setting* `type = "response"` *returns the predicted probabilities rather than predicted log-odds*

For exposure status $e$, the standardized hospitalization proportion is calculated as:

$$\hat{P}_e^{std} = \frac{1}{N} \sum_{i=1}^{N} \hat{P}(Y_i = 1 \mid E_i = e, A_i, S_i),$$

where $N$ is the number of cases in the adjusted analytic cohort, $e$ denotes the assigned exposure status (`Yes` or `No`), and $A_i$ and $S_i$ represent the observed age group and sex of case $i$.

Each prodicted porbability is obtained from the fitted logistic regression model and averaged across the same cohort

```
# Predict hospitalization probabilities under each exposure scenario
pred_exposed <- predict(
    logistic_model,
    newdata = everyone_exposed,
    type = "response"
)

pred_unexposed <- predict(
    logistic_model,
    newdata = everyone_unexposed,
    type = "response"
)

# Average predicted probabilities across the adjusted cohort
adjusted_risk_exposed <- mean(pred_exposed)
adjusted_risk_unexposed <- mean(pred_unexposed)

# Summarize standardized hospitalization proportions
standardized_risk_summary <- data.frame(
    `Exposure Scenario` = c(
        "Reported MediCond",
        "No Reported MedCond"
    ),
    `Cases (N)` = format(
        rep(nrow(adjusted_analysis), 2),
        big.mark = ","
```

```
    ),
    `Standardized Proportion` = sprintf(
        "%.4f",
        c(adjusted_risk_exposed, adjusted_risk_unexposed)
    ),
    `Standardized Percentage` = sprintf(
        "%.1f%%",
        c(adjusted_risk_exposed, adjusted_risk_unexposed) * 100
    ),
    check.names = FALSE
)

knitr::kable(
    standardized_risk_summary,
    caption = "Regression-Standardized Hospitalization Proportions",
    align = c("l", rep("r", ncol(standardized_risk_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 26: Regression-Standardized Hospitalization Proportions

| Exposure Scenario | Cases (N) | Standardized Proportion | Standardized Percentage |
|---|---|---|---|
| Reported MediCond | 1,462 | 0.2551 | 25.5% |
| No Reported MedCond | 1,462 | 0.0669 | 6.7% |

The standardized hospitalization proportion under reported pre-existing medical conditions was 25.5%, compared with 6.7% under no reported medical conditions.

These are model-based estimates averaged across the observed age and sex distribution of the adjusted-analysis cohort, rather than the directly observed hospitalization proportions within each exposure group. Unlike the crude hospitalization proportions, which compare the observed exposure groups with their potentially different demographic compositions, regression standardization evaluates both exposure scenarios using the same age and sex distribution.

## 5.4 Adjusted Risk Ratio and Risk Difference

The regression-standardized hospitalization proportions were used to calculate two measures of adjusted association: the risk ratio (RR) and risk difference (RD). Both measures compare

model-predicted hospitalization proportions under the two exposure scenarios, standardized to the same age and sex distribution.

The adjusted risk ratio compares the relative difference between the standardized hospitalization proportions:

$$RR_{\text{adjusted}} = \frac{\bar{p}E = 1}{\bar{p}E = 0},$$

where $\bar{p}_{E=1}$ and $\bar{p}_{E=0}$ represent the mean model-predicted hospitalization probabilities under reported pre-existing medical conditions and no reported medical conditions, respectively.

The adjusted risk difference expresses the absolute difference between these proportions:

$$_{R}D_{\text{adjusted}} = \bar{p}_{E=1} - \bar{p}_{E=0}.$$

Unlike the risk ratio, which is expressed as a relative measure, the risk difference can be multiplied by 100 and interpreted in percentage points.

### 5.4.1 Estimation of Adjusted Association Measures

```
# Calculate adjusted risk ratio and risk difference
adjusted_rr <- adjusted_risk_exposed / adjusted_risk_unexposed
adjusted_rd <- adjusted_risk_exposed - adjusted_risk_unexposed

# Summarize standardized association measures
adjusted_measures_summary <- data.frame(
    `Measure` = c(
        "Adjusted Risk Ratio",
        "Adjusted Risk Difference",
        "Adjusted Risk Difference (pp)"
    ),
    `Estimate` = c(
        sprintf("%.3f", adjusted_rr),
        sprintf("%.4f", adjusted_rd),
        sprintf("%.2f", adjusted_rd * 100)
    ),
    `Interpretation Scale` = c(
        "Relative",
        "Absolute Proportion",
        "Percentage Points"
```

```
    ),
    check.names = FALSE
)

knitr::kable(
    adjusted_measures_summary,
    caption = "Regression-Standardized Association Measures",
    align = c("l", rep("r", ncol(adjusted_measures_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 27: Regression-Standardized Association Measures

| Measure | Estimate | Interpretation Scale |
|---|---|---|
| Adjusted Risk Ratio | 3.816 | Relative |
| Adjusted Risk Difference | 0.1882 | Absolute Proportion |
| Adjusted Risk Difference (pp) | 18.82 | Percentage Points |

### 5.4.2 Interpretation of Adjusted Risk Ratio

The regression-standardized adjusted risk ratio was 3.82. This indicates that the model-estimated hospitalization proportion under reported pre-existing medical conditions was approximately 3.82 times the corresponding proportion under no reported medical conditions, after standardizing to the same age and sex distribution.

The adjusted risk ratio therefore summarizes the relative association between reported pre-existing medical conditions and hospitalization, accounting for differences in the age and sex distributions of the observed exposure groups.

### 5.4.3 Interpretation of Adjusted Risk Difference

The regression-standardized adjusted risk difference was 0.1882, equivalent to 18.82 percentage points.

This represents the absolute difference between the two standardized hospitalization proportions. In other words, the model estimated approximately 18.8 additional hospitalizations per 100 cases under the reported-medical-conditions scenario compared with the no-reported-medical-conditions scenario, when both scenarios were standardized to the same age and sex distribution.

# 6 Statistical Uncertainty and Bootstrap Inference

The nonparametric bootstrap is a resampling method used to approximate the sampling distribution of a statistic. Rather than repeatedly collecting new datasets, the bootstrap generates samples from the observed data by randomly selecting observations with replacement.

In this analysis, bootstrap resampling is used to evaluate uncertainty in the hospitalization proportion and the regression-standardized measures of association.

## 6.1 Resampling the Analytic Cohort

Each bootstrap sample contains the same number of observations as the original analytic cohort. However, because sampling occurs with replacement, some observations appear multiple times while others are not selected.

Maintaining the original sample size allows the bootstrap procedure to approximate the variability of an estimator across repeated samples of the same size, using the empirical distribution of the observed cohort as the resampling distribution.

### 6.1.1 Constructing a Bootstrap Sample

```
# Set random seed for reproducibility
set.seed(42)

# Number of observations in adjusted analytic cohort
n <- nrow(adjusted_analysis)

# Draw n row indices with replacement
bootstrap_indices <- sample(
    x = seq_len(n),
    size = n,
    replace = TRUE
)

# Construct a bootstrap sample
bootstrap_sample <- adjusted_analysis[
    bootstrap_indices,
    ,
    drop = FALSE
]
```

```r
# Summarize sample sizes
bootstrap_size_summary <- data.frame(
    `Dataset` = c(
        "Original Adjusted Cohort",
        "Bootstrap Sample"
    ),
    `Observations (N)` = format(
        c(nrow(adjusted_analysis), nrow(bootstrap_sample)),
        big.mark = ","
    ),
    check.names = FALSE
)

knitr::kable(
    bootstrap_size_summary,
    caption = "Bootstrap Sample Size Comparison",
    align = c("l", rep("r", ncol(bootstrap_size_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 28: Bootstrap Sample Size Comparison

| Dataset | Observations (N) |
|---|---|
| Original Adjusted Cohort | 1,462 |
| Bootstrap Sample | 1,462 |

The original adjusted analytic cohort contained 1,462 observations. The bootstrap sample also contained 1,462 observations, although the composition of the sample differed because individual cases could be selected more than once.

### 6.1.2 Observation Selection and Sample Composition

For a dataset containing $n$ observations, the porbability that a particular observation is not selected in a single draw is:

$$P(\text{not selected in one draw}) = 1 - \frac{1}{n}.$$

Since a bootstrap sample consists of $n$ independent draws with replacement, the probability that a particular observation is never selected is:

$$P(\text{never selected}) = \left(1 - \frac{1}{n}\right)^n.$$

As $n$ becomes large, this probability approaches:

$$\lim_{n \to \infty} \left(1 - \frac{1}{n}\right)^n = e^{-1} \approx 0.368.$$

As a result, the probability that an observation appears at least once in a bootstrap sample approaches:

$$P(\text{selected at least once}) = 1 - e^{-1} \approx 0.632.$$

A typical bootstrap sample contains approximately 63.2% of the original observations as distinct cases, with repeated selections accounting for the remaining positions.

```
# Count how many times each original observation was selected
selection_counts <- tabulate(
    bootstrap_indices,
    nbins = n
)

# Calculate the number and percentage of unique observations
n_unique <- sum(selection_counts > 0)
n_not_selected <- sum(selection_counts == 0)

prop_unique <- (n_unique / n) * 100
prop_not_selected <- (n_not_selected / n) * 100

# Summarize bootstrap sample composition
selection_summary <- data.frame(
    `Measure` = c(
        "Original Observations",
        "Distinct Observations Selected",
        "Observations Not Selected",
        "Percentage Selected at Least Once",
        "Percentage Not Selected"
    ),
    `Value` = c(
        format(n, big.mark = ","),
        format(n_unique, big.mark = ","),
        format(n_not_selected, big.mark = ","),
```

```
        sprintf("%.1f%%", prop_unique),
        sprintf("%.1f%%", prop_not_selected)
    ),
    check.names = FALSE
)

knitr::kable(
    selection_summary,
    caption = "Bootstrap Sample Composition",
    align = c("l", rep("r", ncol(selection_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 29: Bootstrap Sample Composition

| Measure | Value |
|---|---|
| Original Observations | 1,462 |
| Distinct Observations Selected | 949 |
| Observations Not Selected | 513 |
| Percentage Selected at Least Once | 64.9% |
| Percentage Not Selected | 35.1% |

```
# Summarize the frequency of repeated selections
selection_frequency <- as.data.frame(
    table(selection_counts)
)

names(selection_frequency) <- c(
    "Times_Selected",
    "Number_of_Observations"
)

selection_frequency$Times_Selected <- as.integer(
    as.character(selection_frequency$Times_Selected)
)

selection_frequency$Percentage <- (
    selection_frequency$Number_of_Observations / n
) * 100
```

```r
selection_frequency_summary <- data.frame(
    `Times Selected` = selection_frequency$Times_Selected,
    `Original Observations (n)` = format(
        selection_frequency$Number_of_Observations,
        big.mark = ","
    ),
    `Percentage of Original Cohort` = sprintf(
        "%.1f%%",
        selection_frequency$Percentage
    ),
    check.names = FALSE
)

knitr::kable(
    selection_frequency_summary,
    caption = "Frequency of Observation Selection",
    align = c("l", rep("r", ncol(selection_frequency_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 30: Frequency of Observation Selection

| Times Selected | Original Observations (n) | Percentage of Original Cohort |
|---|---|---|
| 0 | 513 | 35.1% |
| 1 | 564 | 38.6% |
| 2 | 277 | 18.9% |
| 3 | 90 | 6.2% |
| 4 | 16 | 1.1% |
| 5 | 2 | 0.1% |

In this bootstrap sample, 949 distinct observations were selected, representing 64.9% of the original adjusted analytic cohort. The remaining 513 observations (35.1%) were not selected.

### 6.1.3 Comparing Hospitalization Distributions

bootstrap sampling changes the composition of the analytic cohort, the proportion hospitalized may differ between the original dataset and a particular bootstrap sample.

The hospitalization distribution was compared between the original adjusted cohort and the bootstrap sample to illustrate this sampling variability.

```r
# Calculate hospitalization counts
original_hosp <- table(
    adjusted_analysis$hospitalized
)

bootstrap_hosp <- table(
    bootstrap_sample$hospitalized
)

# Calculate hospitalization percentages
original_hosp_pct <- prop.table(original_hosp) * 100
bootstrap_hosp_pct <- prop.table(bootstrap_hosp) * 100

# Construct comparison table
hospitalization_comparison <- data.frame(
    `Hospitalization Status` = names(original_hosp),
    `Original (n)` = as.integer(original_hosp),
    `Original (%)` = sprintf(
        "%.1f%%",
        original_hosp_pct
    ),
    `Bootstrap (n)` = as.integer(bootstrap_hosp),
    `Bootstrap (%)` = sprintf(
        "%.1f%%",
        bootstrap_hosp_pct
    ),
    check.names = FALSE
)

knitr::kable(
    hospitalization_comparison,
    caption = "Hospitalization Distribution: Original vs. Bootstrap Sample",
    align = c("l", rep("r", ncol(hospitalization_comparison) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 31: Hospitalization Distribution: Original vs. Bootstrap Sample

| Hospitalization Status | Original (n) | Original (%) | Bootstrap (n) | Bootstrap (%) |
|---|---|---|---|---|
| No | 1185 | 81.1% | 1197 | 81.9% |

| Hospitalization Status | Original (n) | Original (%) | Bootstrap (n) | Bootstrap (%) |
|---|---|---|---|---|
| Yes | 277 | 18.9% | 265 | 18.1% |

```
# Extract overall hospitalization proportions
original_hosp_risk <- unname(original_hosp_pct["Yes"])
bootstrap_hosp_risk <- unname(bootstrap_hosp_pct["Yes"])

# Calculate percentage-point difference
hosp_difference <- bootstrap_hosp_risk - original_hosp_risk

hospitalization_summary <- data.frame(
    `Measure` = c(
        "Original Hospitalization Proportion",
        "Bootstrap Hospitalization Proportion",
        "Difference (Percentage Points)"
    ),
    `Estimate` = c(
        sprintf("%.2f%%", original_hosp_risk),
        sprintf("%.2f%%", bootstrap_hosp_risk),
        sprintf("%+.2f", hosp_difference)
    ),
    check.names = FALSE
)

knitr::kable(
    hospitalization_summary,
    caption = "Hospitalization Proportion Comparison",
    align = c("l", rep("r", ncol(hospitalization_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 32: Hospitalization Proportion Comparison

| Measure | Estimate |
|---|---|
| Original Hospitalization Proportion | 18.95% |
| Bootstrap Hospitalization Proportion | 18.13% |
| Difference (Percentage Points) | -0.82 |

The observed hospitalization proportion in the original adjusted analytic cohort was 18.9%, compared with 18.1% in the bootstrap sample.

The difference was -0.82 percentage points, reflecting the change in sample composition introduced by resampling with replacement.

This comparison illustrates the variability that can arise from a single bootstrap resample. However, one resample is insufficient to characterize the sampling distribution of the hospitalization proportion. Repeating the procedure many times allows the distribution of bootstrap estimates to be examined and used for statistical inference.

### 6.1.4 Constructing a Bootstrap Distribution

The single bootstrap sample above demonstrates how sampling with replacement changes the composition of the analytic cohort. However, estimating sampling variability requires repeated resampling and recalculation of the statistic of interest for each bootstrap sample.

We first examine the overall hospitalization proportion in the adjusted analytic cohort. The observed hospitalization proportion is defined as:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^{n} Y_i,$$

where $Y_i = 1$ indicates hospitalization for case $i$, $Y_i = 0$ indicates no hospitalization, and $n$ represents the number of cases in the adjusted analytic cohort.

For each bootstrap replication $b$, a sample of $n$ cases is drawn with replacement and the hospitalization proportion is recalculated, producing $\hat{p}^{*(b)}$. Repeating this process $B$ times generates an empirical bootstrap distribution of the estimator.

```
# Calculate original hospitalization proportion
original_prop <- mean(
    adjusted_analysis$hospitalized == "Yes"
)

# Set random seed for reproducibility
set.seed(42)

# Number of bootstrap replications
B <- 1000

# Number of cases in adjusted analytic cohort
n <- nrow(adjusted_analysis)

# Initialize vector to store bootstrap proportions
bootstrap_props <- numeric(B)
```

```r
# Generate bootstrap distribution
for (b in seq_len(B)) {

    # Sample case indices with replacement
    indices <- sample(
        x = seq_len(n),
        size = n,
        replace = TRUE
    )

    # Construct bootstrap sample
    boot_data <- adjusted_analysis[
        indices,
        ,
        drop = FALSE
    ]

    # Calculate hospitalization proportion
    bootstrap_props[b] <- mean(
        boot_data$hospitalized == "Yes"
    )
}

# Summarize original and bootstrap estimates
bootstrap_distribution_summary <- data.frame(
    `Measure` = c(
        "Original Hospitalization Proportion",
        "Mean Bootstrap Proportion",
        "Minimum Bootstrap Proportion",
        "Maximum Bootstrap Proportion",
        "Number of Replications"
    ),
    `Value` = c(
        sprintf("%.4f", original_prop),
        sprintf("%.4f", mean(bootstrap_props)),
        sprintf("%.4f", min(bootstrap_props)),
        sprintf("%.4f", max(bootstrap_props)),
        format(B, big.mark = ",")
    ),
    check.names = FALSE
)
```

```
knitr::kable(
    bootstrap_distribution_summary,
    caption = "Bootstrap Distribution of Hospitalization Proportion",
    align = c("l", rep("r", ncol(bootstrap_distribution_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 33: Bootstrap Distribution of Hospitalization Proportion

| Measure | Value |
|---|---|
| Original Hospitalization Proportion | 0.1895 |
| Mean Bootstrap Proportion | 0.1895 |
| Minimum Bootstrap Proportion | 0.1587 |
| Maximum Bootstrap Proportion | 0.2257 |
| Number of Replications | 1,000 |

The original hospitalization proportion was 18.95%. Across 1,000 bootstrap replications, the mean estimated hospitalization proportion was 18.95%.

The bootstrap estimates provide an empirical approximation to the sampling distribution of the hospitalization proportion under independent case-level resampling. Examining the center, spread, and shape of this distribution helps characterize the estimator's sampling variability.

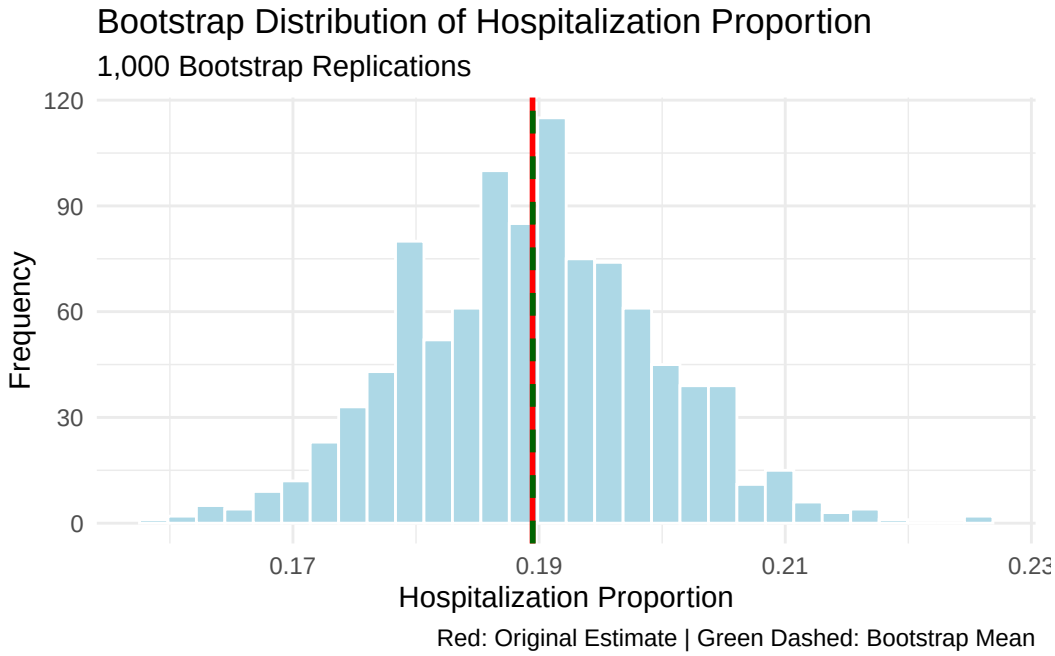
#### 6.1.4.1 Visualizing the Bootstrap Distribution

The histogram below displays the distribution of hospitalization proportions obtained from the bootstrap replications. The solid red line represents the original observed hospitalization proportion, while the dashed green line represents the mean of the bootstrap estimates.

```
# Convert bootstrap estimates to a data frame
bootstrap_prop_df <- data.frame(
    Hospitalization_Proportion = bootstrap_props
)

# Plot bootstrap distribution
ggplot(
    bootstrap_prop_df,
    aes(x = Hospitalization_Proportion)
) +
    geom_histogram(
```

```
        bins = 30,
        fill = "lightblue",
        color = "white"
    ) +
    geom_vline(
        xintercept = original_prop,
        color = "red",
        linewidth = 1
    ) +
    geom_vline(
        xintercept = mean(bootstrap_props),
        color = "darkgreen",
        linewidth = 1,
        linetype = "dashed"
    ) +
    labs(
        title = "Bootstrap Distribution of Hospitalization Proportion",
        subtitle = paste(
            format(B, big.mark = ","),
            "Bootstrap Replications"
        ),
        x = "Hospitalization Proportion",
        y = "Frequency",
        caption = paste(
            "Red: Original Estimate |",
            "Green Dashed: Bootstrap Mean"
        )
    ) +
    theme_minimal()
```




The bootstrap distribution is centered near the original observed hospitalization proportion. The mean bootstrap estimate was 0.1895, compared with the original estimate of 0.1895.

The difference between the bootstrap mean and the original estimate was +0.0000, indicating that the bootstrap estimates were centered close to the observed statistic.

*The bootstrap distribution demonstrates that the estimated hospitalization proportion varies across samples of the same size, even when all samples are drawn from the same observed cohort.*

#### 6.1.4.2 Bootstrap Standard Error

The bootstrap standard error is calculated as the sample standard deviation of the $B$ bootstrap estimates:

$$\widehat{\mathrm{SE}}_{\mathrm{boot}}(\hat{p}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{p}^{*(b)} - \bar{p}^{*}\right)^2}.$$

Here, $\hat{p}^{*(b)}$ represents the hospitalization proportion estimated in bootstrap replication $b$, and $\bar{p}^{*}$ represents the mean of the bootstrap estimates.

Unlike the standard deviation of individual hospitalization outcomes, which describes variation across cases, the bootstrap standard error describes variation in the estimated hospitalization proportion across resamples.

#### 6.1.4.3 Percentile Bootstrap Confidence Interval

A 95% percentile bootstrap confidence interval was constructed using the 2.5th and 97.5th percentiles of the bootstrap distribution:

$$CI_{95\%} = [Q_{0.025}(\hat{p}^*), Q_{0.975}(\hat{p}^*)] \,.$$

This approach uses the empirical distribution of bootstrap estimates to construct an interval without requiring an exactly normal sampling distribution.

However, percentile confidence intervals still depend on the bootstrap distribution adequately approximating the estimator's sampling behavior.

```
# Calculate bootstrap standard error
bootstrap_se <- sd(bootstrap_props)

# Calculate 95% percentile bootstrap confidence interval
bootstrap_ci <- quantile(
    bootstrap_props,
    probs = c(0.025, 0.975)
)

# Calculate bootstrap mean
bootstrap_mean <- mean(bootstrap_props)

# Construct formatted inference summary
bootstrap_summary <- data.frame(
    `Statistic` = c(
        "Original Hospitalization Proportion",
        "Mean Bootstrap Proportion",
        "Bootstrap Standard Error",
        "95% CI Lower Bound",
        "95% CI Upper Bound"
    ),
    `Estimate` = sprintf(
        "%.4f",
        c(
            original_prop,
            bootstrap_mean,
            bootstrap_se,
            bootstrap_ci[1],
            bootstrap_ci[2]
        )
```

```
    ),
    `Percentage Scale` = sprintf(
        "%.2f%%",
        c(
            original_prop,
            bootstrap_mean,
            bootstrap_se,
            bootstrap_ci[1],
            bootstrap_ci[2]
        ) * 100
    ),
    check.names = FALSE
)

knitr::kable(
    bootstrap_summary,
    caption = "Bootstrap Inference for Hopitalization Proportion",
    align = c("l", rep("r", ncol(bootstrap_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 34: Bootstrap Inference for Hopitalization Proportion

| Statistic | Estimate | Percentage Scale |
|---|---|---|
| Original Hospitalization Proportion | 0.1895 | 18.95% |
| Mean Bootstrap Proportion | 0.1895 | 18.95% |
| Bootstrap Standard Error | 0.0101 | 1.01% |
| 95% CI Lower Bound | 0.1703 | 17.03% |
| 95% CI Upper Bound | 0.2086 | 20.86% |

The observed hospitalization proportion in the adjusted analytic cohort was 18.95%. Across 1,000 bootstrap replications, the mean estimated hospitalization proportion was 18.95%, with a bootstrap standard error of 0.0101, equivalent to 1.01 percentage points.

## 6.2 Refitting the Regression Model

Unlike the overall hospitalization proportion, the regression-standardized estimates depend on coefficients estimated from the observed data. If the composition of the analytic cohort changes, the fitted regression coefficients may also change.

As a result, each bootstrap replication must repeat the model-fitting procedure rather than reuse the coefficients estimated from the original analytic cohort.

For each bootstrap replication, a sample of $n$ case records is drawn with replacement, and the same multivariable logistic regression model is refitted. Repeating this procedure produces bootstrap distributions of the estimated regression coefficients.

The resulting distribution provides an empirical approximation to the sampling variability of the estimated log-odds ratio under the assumed case-level resampling procedure.

### 6.2.1 Generating Bootstrap Regression Models

```
# Set random seed for reproducibility
set.seed(42)

# Number of bootstrap replications
B_model <- 2000

# Number of observations in adjusted analytic cohort
n <- nrow(adjusted_analysis)

# Initialize storage for bootstrap coefficients
bootstrap_beta <- rep(NA_real_, B_model)

# Initialize storage for fitted models and bootstrap samples
bootstrap_models <- vector("list", B_model)
bootstrap_samples <- vector("list", B_model)

# Refit the regression model for each bootstrap replication
for (b in seq_len(B_model)) {

    # Sample case indices with replacement
    indices <- sample(
        x = seq_len(n),
        size = n,
        replace = TRUE
    )

    # Construct bootstrap dataset
    boot_data <- adjusted_analysis[
        indices,
        ,
```

```
        drop = FALSE
    ]

    # Refit logistic regression model
    boot_model <- tryCatch(
        suppressWarnings(
            glm(
                hospitalized ~ medcond + age_factor + sex_factor,
                data = boot_data,
                family = binomial(link = "logit")
            )
        ),
        error = function(e) NULL
    )

    # Skip unsuccessful model fits
    if (is.null(boot_model)) {
        next
    }

    # Verify convergence and finite coefficients
    valid_fit <- isTRUE(boot_model$converged) &&
        exposure_term %in% names(coef(boot_model)) &&
        all(is.finite(coef(boot_model)))

    if (valid_fit) {

        # Store exposure coefficient
        bootstrap_beta[b] <- unname(
            coef(boot_model)[exposure_term]
        )

        # Store fitted model and corresponding bootstrap sample
        bootstrap_models[[b]] <- boot_model
        bootstrap_samples[[b]] <- boot_data
    }
}

# Identify successful bootstrap replications
valid_beta <- bootstrap_beta[is.finite(bootstrap_beta)]

n_successful <- length(valid_beta)
```

```r
n_failed <- B_model - n_successful

# Summarize bootstrap model fitting
model_fit_summary <- data.frame(
    `Measure` = c(
        "Bootstrap Replications Attempted",
        "Successful Model Fits",
        "Unsuccessful Model Fits",
        "Successful Replications (%)"
    ),
    `Value` = c(
        format(B_model, big.mark = ","),
        format(n_successful, big.mark = ","),
        format(n_failed, big.mark = ","),
        sprintf("%.1f%%", 100 * n_successful / B_model)
    ),
    check.names = FALSE
)

knitr::kable(
    model_fit_summary,
    caption = "Bootstrap Regression Model Fitting Summary",
    align = c("l", rep("r", ncol(model_fit_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 35: Bootstrap Regression Model Fitting Summary

| Measure | Value |
|---|---|
| Bootstrap Replications Attempted | 2,000 |
| Successful Model Fits | 2,000 |
| Unsuccessful Model Fits | 0 |
| Successful Replications (%) | 100.0% |

A total of 2000 bootstrap regression models were attempted. Of these, 2000 produced converged models with finite coefficient estimates, while 0 were unsuccessful.

Each successful replication represents a logistic regression model fitted to a resampled version of the adjusted analytic cohort. Do to the change of observations being selected multiple times or omitted from an individual resample, the fitted coefficients can differ from those estimated using the original dataset.

### 6.2.2 Summarizing the Bootstrap Exposure Coefficients

The exposure coefficient from each successful bootstrap model was retained to examine variability in the estimated conditional log-odds ratio.

```
# Original exposure coefficient
original_beta <- unname(
    coef(logistic_model)[exposure_term]
)

# Calculate bootstrap coefficient summary statistics
beta_mean <- mean(valid_beta)
beta_se <- sd(valid_beta)
beta_median <- median(valid_beta)

# Construct formatted coefficient summary
beta_summary <- data.frame(
    `Statistic` = c(
        "Original Exposure Coefficient",
        "Mean Bootstrap Coefficient",
        "Median Bootstrap Coefficient",
        "Bootstrap Standard Error",
        "Minimum Bootstrap Coefficient",
        "Maximum Bootstrap Coefficient"
    ),
    `Estimate` = sprintf(
        "%.4f",
        c(
            original_beta,
            beta_mean,
            beta_median,
            beta_se,
            min(valid_beta),
            max(valid_beta)
        )
    ),
    check.names = FALSE
)

knitr::kable(
    beta_summary,
    caption = "Bootstrap Distribution of Exposure Coefficients",
    align = c("l", rep("r", ncol(beta_summary) - 1)),
```

```
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 36: Bootstrap Distribution of Exposure Coefficients

| Statistic | Estimate |
|---|---|
| Original Exposure Coefficient | 1.8737 |
| Mean Bootstrap Coefficient | 1.9081 |
| Median Bootstrap Coefficient | 1.9013 |
| Bootstrap Standard Error | 0.2400 |
| Minimum Bootstrap Coefficient | 1.1574 |
| Maximum Bootstrap Coefficient | 2.9253 |

The exposure coefficient estimated from the original logistic regression model was 1.874. Across the 2000 successful bootstrap replications, the mean exposure coefficient was 1.908, with a bootstrap standard error of 0.240.

The bootstrap standard error describes the variability of the estimated exposure coefficient across resampled datasets. Unlike the model-based standard error obtained from the original logistic regression, this estimate is calculated from the empirical distribution of bootstrap coefficients.

### 6.2.3 Examining the Bootstrap Coefficient Distribution

The bootstrap distribution of the medical-condition coefficient was examined to assess its variability across resampled analytic cohorts. The logistic regression estimates coefficients are on the log-odds scale, so this distribution represents the sampling variability of the estimated conditional log-odds ratio.

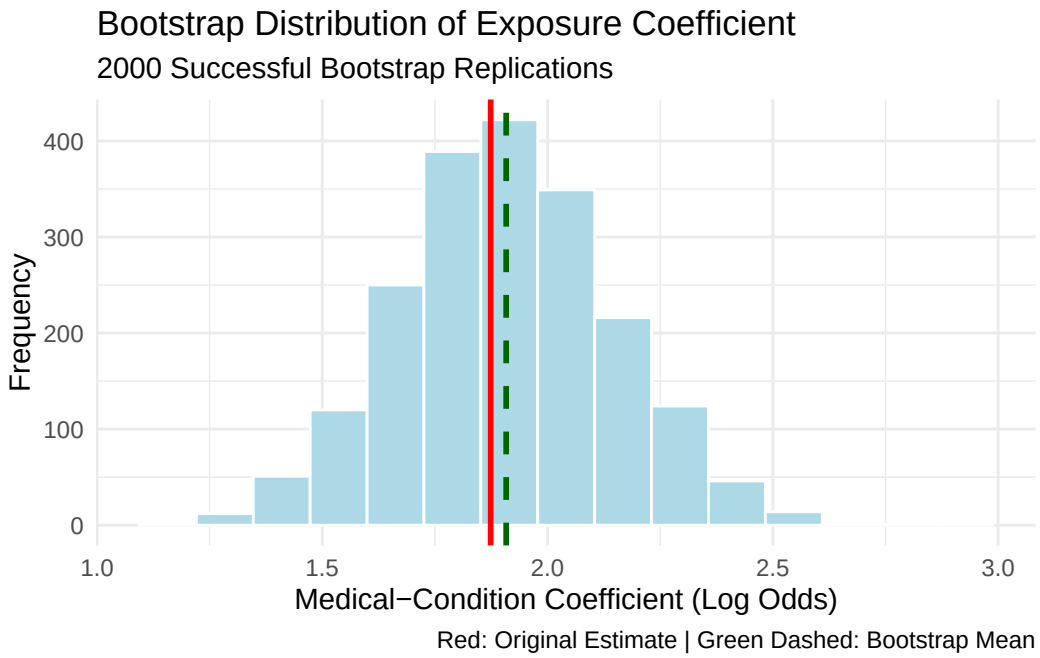
The histogram below displays the exposure coefficients from successful bootstrap replications. The solid red line represents the coefficient estimated from the original analytic cohort, while the dashed green line represents the mean of the bootstrap estimates.

```
# Retain successful bootstrap coefficient estimates
valid_beta <- bootstrap_beta[is.finite(bootstrap_beta)]

# Original exposure coefficient
original_beta <- unname(
    coef(logistic_model)[exposure_term]
)
```

```r
# Construct data frame for visualization
bootstrap_beta_df <- data.frame(
    Exposure_Coefficient = valid_beta
)

# Plot bootstrap coefficient distribution
ggplot(
    bootstrap_beta_df,
    aes(x = Exposure_Coefficient)
) +
    geom_histogram(
        bins = 15,
        fill = "lightblue",
        color = "white"
    ) +
    geom_vline(
        xintercept = original_beta,
        color = "red",
        linewidth = 1
    ) +
    geom_vline(
        xintercept = mean(valid_beta),
        color = "darkgreen",
        linewidth = 1,
        linetype = "dashed"
    ) +
    labs(
        title = "Bootstrap Distribution of Exposure Coefficient",
        subtitle = paste(
            length(valid_beta),
            "Successful Bootstrap Replications"
        ),
        x = "Medical-Condition Coefficient (Log Odds)",
        y = "Frequency",
        caption = paste(
            "Red: Original Estimate |",
            "Green Dashed: Bootstrap Mean"
        )
    ) +
    theme_minimal()
```




The medical-condition coefficient estimated from the original logistic regression model was 1.874 on the log-odds scale. Across 2000 successful bootstrap replications, the mean coefficient was 1.908, with a bootstrap standard error of 0.240.

The bootstrap coefficients varied across resamples, illustrating the sensitivity of the estimated conditional association between reported pre-existing medical conditions and hospitalization to the composition of the analytic cohort.

### 6.2.4 Recalculating Standardized Estimates

To extend bootstrap inference to the adjusted risk ratio (RR) and adjusted risk difference (RD), regression standardization was repeated for each successfully fitted bootstrap model.

For each replication, two exposure scenarios were constructed using the corresponding bootstrap sample. Medical-condition status was set to `Yes` for all cases in one scenario and `No` for all cases in the other, while age group and sex remained unchanged.

Predicted hospitalization probabilities were then generated from the corresponding bootstrap regression model and averaged across the bootstrap sample.

For bootstrap replication $b$, the standardized hospitalization proportion under exposure condition $e$ is:

$$\hat{p}_e^{*(b)} = \frac{1}{n} \sum_{i=1}^{n} \hat{P}^{*(b)} \left( Y_i = 1 \mid E_i = e, A_i^{*(b)}, S_i^{*(b)} \right).$$

Here, $\hat{P}^{*(b)}$ denotes the predicted hospitalization probability from the regression model fitted in bootstrap replication $b$, and $A_i^{*(b)}$ and $S_i^{*(b)}$ represent the age group and sex of the $i$th observation in that bootstrap sample.

The adjusted risk ratio and risk difference are recalculated as:

$$\widehat{RR}^{*(b)} = \frac{\hat{p}_1^{*(b)}}{\hat{p}_0^{*(b)}},$$

$$\widehat{RD}^{*(b)} = \hat{p}_1^{*(b)} - \hat{p}_0^{*(b)}.$$

Repeating this procedure across bootstrap models produces empirical distributions of the standardized measures of association.

### 6.2.5 Calculating Standardized Estimates

```
# Initialize storage for standardized bootstrap estimates
bootstrap_rr <- rep(NA_real_, B_model)
bootstrap_rd <- rep(NA_real_, B_model)

bootstrap_risk_exposed <- rep(NA_real_, B_model)
bootstrap_risk_unexposed <- rep(NA_real_, B_model)

# Apply regression standardization to stored bootstrap models
for (b in seq_len(B_model)) {

    # Retrieve fitted model and corresponding bootstrap sample
    boot_model <- bootstrap_models[[b]]
    boot_data <- bootstrap_samples[[b]]

    # Skip unsuccessful model fits
    if (is.null(boot_model) || is.null(boot_data)) {
        next
    }

    # Construct exposure scenarios
    exposed <- boot_data
    unexposed <- boot_data

    # Assign exposure status while preserving factor levels
```

```
    exposed$medcond <- factor(
        rep("Yes", nrow(boot_data)),
        levels = levels(adjusted_analysis$medcond)
    )

    unexposed$medcond <- factor(
        rep("No", nrow(boot_data)),
        levels = levels(adjusted_analysis$medcond)
    )

    # Predict hospitalization probabilities
    predictions <- tryCatch(
        list(
            exposed = predict(
                boot_model,
                newdata = exposed,
                type = "response"
            ),
            unexposed = predict(
                boot_model,
                newdata = unexposed,
                type = "response"
            )
        ),
        error = function(e) NULL
    )

    # Skip unsuccessful predictions
    if (is.null(predictions)) {
        next
    }

    # Standardize over the corresponding bootstrap sample
    risk_exposed <- mean(predictions$exposed)
    risk_unexposed <- mean(predictions$unexposed)

    # Verify finite, valid probabilities
    valid_risks <- is.finite(risk_exposed) &&
        is.finite(risk_unexposed) &&
        risk_exposed >= 0 &&
        risk_exposed <= 1 &&
        risk_unexposed > 0 &&
```

```
        risk_unexposed <= 1

    if (!valid_risks) {
        next
    }

    # Store standardized hospitalization proportions
    bootstrap_risk_exposed[b] <- risk_exposed
    bootstrap_risk_unexposed[b] <- risk_unexposed

    # Calculate standardized association measures
    bootstrap_rr[b] <- risk_exposed / risk_unexposed
    bootstrap_rd[b] <- risk_exposed - risk_unexposed
}

# Identify replications with valid estimates for both measures
valid_replicates <- is.finite(bootstrap_rr) &
    is.finite(bootstrap_rd)

n_valid_standardized <- sum(valid_replicates)
n_failed_standardized <- B_model - n_valid_standardized

# Summarize successful standardization replications
standardization_summary <- data.frame(
    `Measure` = c(
        "Bootstrap Replications Attempted",
        "Valid Standardized Estimates",
        "Unsuccessful Replications",
        "Successful Replications (%)"
    ),
    `Value` = c(
        format(B_model, big.mark = ","),
        format(n_valid_standardized, big.mark = ","),
        format(n_failed_standardized, big.mark = ","),
        sprintf(
            "%.1f%%",
            100 * n_valid_standardized / B_model
        )
    ),
    check.names = FALSE
)
```

```
knitr::kable(
    standardization_summary,
    caption = "Bootstrap Regression Standardization Summary",
    align = c("l", rep("r", ncol(standardization_summary) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 37: Bootstrap Regression Standardization Summary

| Measure | Value |
|---|---|
| Bootstrap Replications Attempted | 2,000 |
| Valid Standardized Estimates | 2,000 |
| Unsuccessful Replications | 0 |
| Successful Replications (%) | 100.0% |

Of the 2000 attempted bootstrap replications, 2000 produced valid standardized hospitalization proportions and corresponding adjusted risk ratios and risk differences.

Each successful replication used a newly fitted logistic regression model and the age and sex distribution of its corresponding bootstrap sample. Consequently, variation in the resulting estimates reflects both changes in the fitted regression coefficients and changes in the empirical distribution of the adjustment variables.

### 6.2.6 Examining the Bootstrap Distributions

The bootstrap distributions of the adjusted risk ratio and risk difference were examined to assess their centers, spreads, and shapes.

Unlike the distribution of the exposure coefficient, which describes variability on the conditional log-odds scale, these distributions describe variability in the regression-standardized measures of association.

The red reference lines represent the adjusted estimates obtained from the original analytic cohort, while the dashed green lines represent the means of the corresponding bootstrap distributions.

```
# Retain successful standardized estimates
valid_rr <- bootstrap_rr[valid_replicates]
valid_rd <- bootstrap_rd[valid_replicates]
```

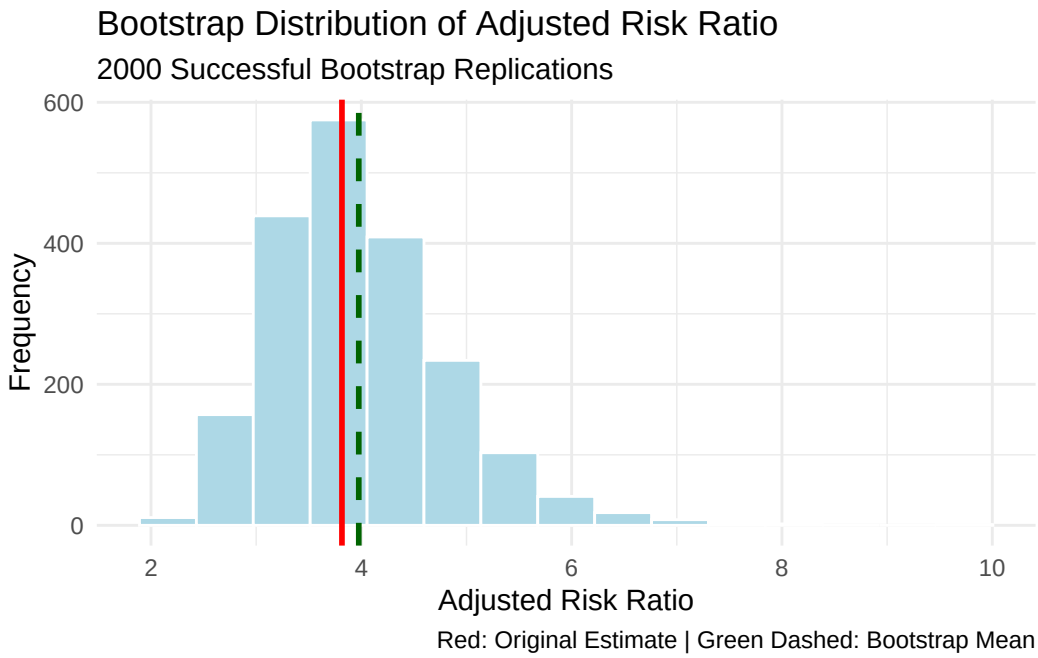
```r
# Require sufficient valid replications
stopifnot(length(valid_rr) >= 2)

# Construct data frames for visualization
bootstrap_rr_df <- data.frame(
    Adjusted_RR = valid_rr
)

bootstrap_rd_df <- data.frame(
    Adjusted_RD = valid_rd
)

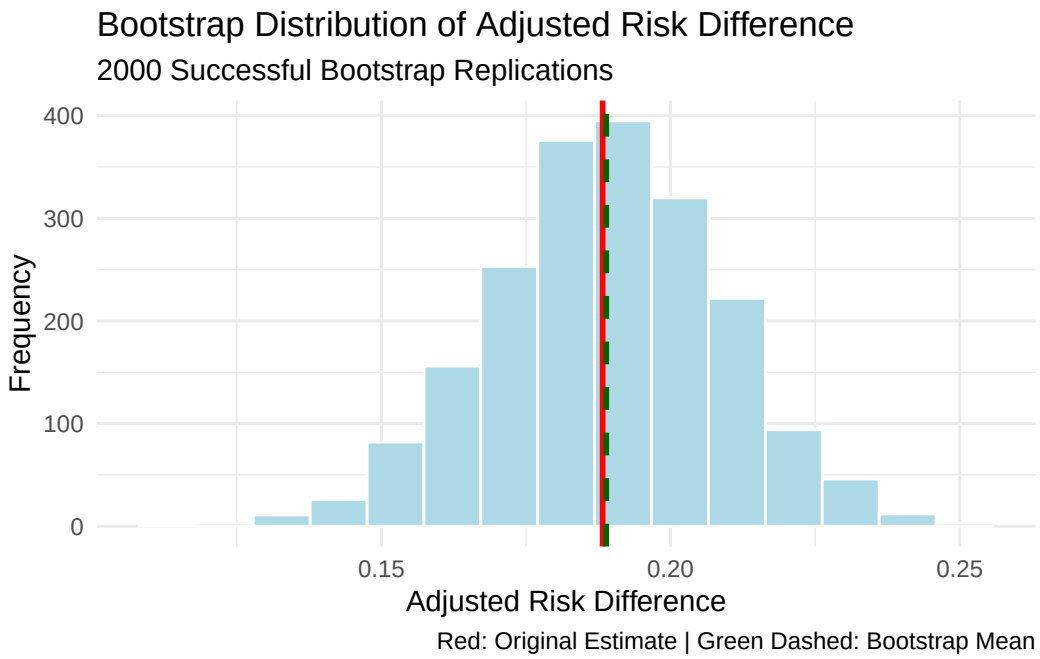
# Plot bootstrap distribution of adjusted risk ratio
ggplot(
    bootstrap_rr_df,
    aes(x = Adjusted_RR)
) +
    geom_histogram(
        bins = 15,
        fill = "lightblue",
        color = "white"
    ) +
    geom_vline(
        xintercept = adjusted_rr,
        color = "red",
        linewidth = 1
    ) +
    geom_vline(
        xintercept = mean(valid_rr),
        color = "darkgreen",
        linewidth = 1,
        linetype = "dashed"
    ) +
    labs(
        title = "Bootstrap Distribution of Adjusted Risk Ratio",
        subtitle = paste(
            length(valid_rr),
            "Successful Bootstrap Replications"
        ),
        x = "Adjusted Risk Ratio",
        y = "Frequency",
        caption = paste(
```

```
            "Red: Original Estimate |",
            "Green Dashed: Bootstrap Mean"
        )
    ) +
    theme_minimal()
```




```
# Plot bootstrap distribution of adjusted risk difference
ggplot(
    bootstrap_rd_df,
    aes(x = Adjusted_RD)
) +
    geom_histogram(
        bins = 15,
        fill = "lightblue",
        color = "white"
    ) +
    geom_vline(
        xintercept = adjusted_rd,
        color = "red",
        linewidth = 1
    ) +
    geom_vline(
        xintercept = mean(valid_rd),
```

```
        color = "darkgreen",
        linewidth = 1,
        linetype = "dashed"
    ) +
    labs(
        title = "Bootstrap Distribution of Adjusted Risk Difference",
        subtitle = paste(
            length(valid_rd),
            "Successful Bootstrap Replications"
        ),
        x = "Adjusted Risk Difference",
        y = "Frequency",
        caption = paste(
            "Red: Original Estimate |",
            "Green Dashed: Bootstrap Mean"
        )
    ) +
    theme_minimal()
```




The bootstrap distributions illustrate the variability of the adjusted association measures across resampled analytic cohorts.

The adjusted risk ratio and risk difference may exhibit different distributional shapes because they summarize associations on different scales. In particular, the risk ratio involves division

by the standardized hospitalization proportion in the unexposed scenario, which can produce a more asymmetric distribution when that denominator varies across resamples.

### 6.2.7 Bootstrap Confidence Intervals

The standard deviation of each bootstrap distribution was used to estimate the corresponding bootstrap standard error.

A 95% percentile bootstrap confidence interval was constructed using the 2.5th and 97.5th percentiles of the successful bootstrap estimates:

$$CI_{95\%}(\hat{\theta}) = \left[Q_{0.025}(\hat{\theta}^*), Q_{0.975}(\hat{\theta}^*)\right],$$

where $\hat{\theta}^*$ represents the bootstrap estimates of the adjusted risk ratio or risk difference.

The original regression-standardized estimates were retained as the primary point estimates, while the bootstrap distributions were used to quantify their uncertainty.

```
# Calculate bootstrap standard errors
se_rr <- sd(valid_rr)
se_rd <- sd(valid_rd)

# Calculate 95% percentile bootstrap confidence intervals
ci_rr <- quantile(
    valid_rr,
    probs = c(0.025, 0.975)
)

ci_rd <- quantile(
    valid_rd,
    probs = c(0.025, 0.975)
)

# Calculate bootstrap means
mean_rr <- mean(valid_rr)
mean_rd <- mean(valid_rd)

# Construct formatted results table
bootstrap_results <- data.frame(
    `Measure` = c(
        "Adjusted RR",
        "Adjusted RD",
```

```
        "Adjusted RD (pp)"
    ),
    `Original Estimate` = c(
        sprintf("%.3f", adjusted_rr),
        sprintf("%.4f", adjusted_rd),
        sprintf("%.2f", adjusted_rd * 100)
    ),
    `Bootstrap Mean` = c(
        sprintf("%.3f", mean_rr),
        sprintf("%.4f", mean_rd),
        sprintf("%.2f", mean_rd * 100)
    ),
    `Bootstrap SE` = c(
        sprintf("%.3f", se_rr),
        sprintf("%.4f", se_rd),
        sprintf("%.2f", se_rd * 100)
    ),
    `95% CI Lower` = c(
        sprintf("%.3f", ci_rr[1]),
        sprintf("%.4f", ci_rd[1]),
        sprintf("%.2f", ci_rd[1] * 100)
    ),
    `95% CI Upper` = c(
        sprintf("%.3f", ci_rr[2]),
        sprintf("%.4f", ci_rd[2]),
        sprintf("%.2f", ci_rd[2] * 100)
    ),
    check.names = FALSE
)

cat("Bootstrap Inference for Regression-Standardized Measures\n")
```

```
Bootstrap Inference for Regression-Standardized Measures
```

```
knitr::kable(
    bootstrap_results,
    caption = "Bootstrap Inference for Regression-Standardized Measures",
    align = c("l", rep("r", ncol(bootstrap_results) - 1)),
    booktabs = TRUE,
    row.names = FALSE
)
```

Table 38: Bootstrap Inference for Regression-Standardized Measures

| Measure | Original Estimate | Bootstrap Mean | Bootstrap SE | 95% CI Lower | 95% CI Upper |
|---|---|---|---|---|---|
| Adjusted RR | 3.816 | 3.976 | 0.836 | 2.664 | 5.914 |
| Adjusted RD | 0.1882 | 0.1889 | 0.0199 | 0.1488 | 0.2278 |
| Adjusted RD (pp) | 18.82 | 18.89 | 1.99 | 14.88 | 22.78 |

The regression-standardized adjusted risk ratio was 3.82 (95% percentile bootstrap CI: 2.66–5.91). This indicates a positive adjusted association between reported pre-existing medical conditions and hospitalization.

The adjusted risk difference was 18.82 percentage points (95% percentile bootstrap CI: 14.88–22.78 percentage points).

Both confidence intervals were derived from the empirical distributions of the regression-standardized estimates. They quantify uncertainty under the assumed case-level resampling procedure.

# 7 Conclusion

This analysis examined the association between reported pre-existing medical conditions and hospitalization among early-pandemic COVID-19 cases using a sample of CDC surveillance data from March 15 through December 10, 2020. Among cases with known exposure and hospitalization status, reported pre-existing medical conditions were associated with a substantially higher proportion of hospitalization. The crude risk ratio was 9.86 (95% CI: 6.72–14.47), with an absolute risk difference of 31.7 percentage points.

Adjustment for age and sex reduced the estimated magnitude of the association. Regression standardization produced an adjusted risk ratio of 3.82 (95% percentile bootstrap CI: 2.66–5.91) and an adjusted risk difference of 18.8 percentage points (95% percentile bootstrap CI: 14.9–22.8 percentage points). The difference between the crude and adjusted estimates illustrates the importance of accounting for potential confounders. These were identified as age and sex variables in the dataset.

Several limitations restrict the interpretation and generalizability of these findings. Only 14.64% of the original surveillance sample had known medical-condition and hospitalization status, which could influence selection bias from complete-case analysis. Additionally, the original sample was constructed using randomly selected reporting dates and contiguous record blocks rather than simple random sampling, limiting its representativeness. The regression model adjusted for age and sex but could not account for other potential confounders, and

the bootstrap confidence intervals reflect uncertainty under independent case-level resampling rather than the original sampling design. Consequently, the results should not be generalized to all reported COVID-19 cases or interpreted as population-level hospitalization risks.